

Exhibit 96

Part 2 of 2

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart

EXHIBIT 96 (Part 2)

Domestic Industry Claim Chart for Claims of U.S. Patent No. 12,373,366 (the “’366 Patent”) Based on Practicing ’366 Products – Part II of II

As shown in the chart below, the Practicing ’366 Products, which include Netlist DDR5 VLP RDIMMs, Netlist Lightning DDR5 RDIMMs, Netlist DDR5 LP MRDIMMs, [REDACTED], and SK hynix RDIMMs and MRDIMMs, practice at least claim 1 of the ’366 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Practicing ’366 Products do not practice the ’366 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Practicing ’366 Products, such differences are insubstantial and the Practicing ’366 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Practicing ’366 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Practicing ’366 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

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Claim 1	Evidence of Use																								
[1.b.2] (cont'd) the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;	Ex. CC-85 (JESD301-1) at 40 (annotated). Table 100 — Register 0x05 <table><tr><th colspan="4">R05</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description^{1,2}</th></tr><tr><td>2:0</td><td>ROE</td><td>0</td><td> R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved at least three bits are written when Vin_Bulk overvoltage is detected </td></tr></table> 1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory. 2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation. 3. This register is set only if PMIC generates internal VR Disable command due to fault condition. Ex. CC-85 (JESD301-1) at 113 (annotated). Table 103 — Register 0x08 <table><tr><th colspan="4">R08</th></tr><tr><th>Bits</th><th>Attribute</th><th>Default</th><th>Description</th></tr><tr><td colspan="4">R08 [7:0]: VIN_BULK Input Power GOOD STATUS</td></tr></table>	R05				Bits	Attribute	Default	Description ^{1,2}	2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved at least three bits are written when Vin_Bulk overvoltage is detected	R08				Bits	Attribute	Default	Description	R08 [7:0]: VIN_BULK Input Power GOOD STATUS			
R05																									
Bits	Attribute	Default	Description ^{1,2}																						
2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved at least three bits are written when Vin_Bulk overvoltage is detected																						
R08																									
Bits	Attribute	Default	Description																						
R08 [7:0]: VIN_BULK Input Power GOOD STATUS																									

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0	RO	0	R08 [0]: VIN_BULK_INPUT_OVER_VOLTAGE_STATUS VIN_Bulk Input Supply Over Voltage Status ⁸ 0 = No Over Voltage 1 = Over Voltage
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8. This register is set when VIN_Bulk input voltage goes above the threshold setting in register Table 122, "Register 0x1B" [7].

Ex. CC-85 (JESD301-1) at 116 (annotated).

See also Ex. CC-82 (JESD301-2) at 46, 53, 55–56, 114, 116, 132; Ex. CC-83 (JESD301-4A) at 30, 35, 38, 111, 114, 133; Ex. CC-84 (JESD301-6) at 50, 57, 59, 125, 127, 143.

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[1.c] a set of components coupled to the PCB,

Each of the Practicing '366 Products includes a set of components coupled to the PCB, such as DDR5 SDRAMs, RCD, PMIC, SPD, and TS, as further explained in element 1[d2].

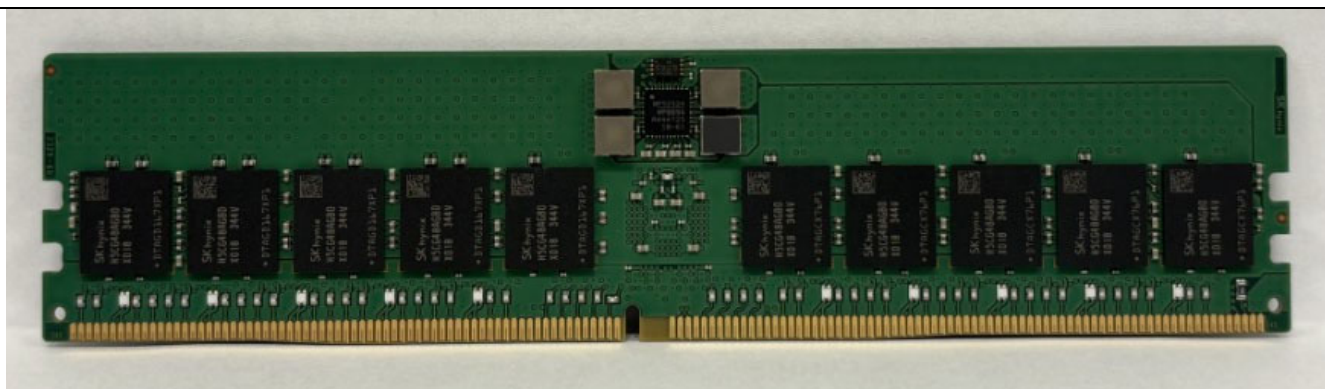
For example, Netlist advertises that “DDR5 DIMMs are the physical modules that contain DDR5 memory chips,” and “DDR5 RDIMM includes a Power Management Integrated Circuit (PMIC).” Ex. CC-33 (Netlist DDR5 DIMMs) at 2. Additionally, Netlist advertises that DDR5 DIMMs include “Temperature Sensor-ICs on DIMM” as a feature. *Id.* at 3.

See also, e.g., id. at 2 (showing, coupled to Netlist DDR5 VLP RDIMM’s PCB, SDRAMs, PMIC, SPD Hub, and TS):



See also images of Netlist RDIMMs (SK hynix 32GB DDR5-5600 RDIMM PC5-44800R Dual Rank x8 Module):

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Ex. CC-36 (Product Images) at 10 (showing, coupled to Netlist DDR5 RDIMM's PCB, SDRAMs, PMIC, TS, and SPD Hub).



Ex. CC-36 (Product Images) at 7 (showing, coupled to Netlist DDR5 RDIMM's PCB, SDRAMs and RCD).

See also images of Netlist Overclock RDIMMs (Netlist 64GB 2Rx4 PC5-6000OC6-R):

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Ex. CC-36 (Product Images) at 3 (showing, coupled to Netlist DDR5 Overclock RDIMM's PCB, SDRAMs, PMIC, TS, and SPD Hub).



Ex. CC-36 (Product Images) at 6 (showing, coupled to Netlist DDR5 Overclock RDIMM's PCB, SDRAMs and RCD).

See also images of Netlist MRDIMMs (SK hynix 32GB DDR5-8800 MCR DIMM Dual Rank x8 Module):

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Ex. CC-36 (Product Images) at 16 (showing, coupled to Netlist DDR5 MRDIMM's PCB, SDRAMs, MDBs, PMIC, TS, and SPD Hub).

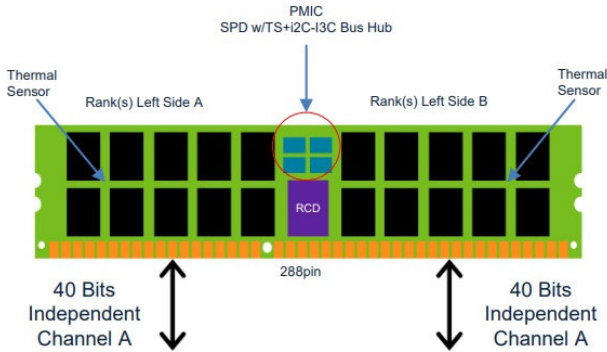


Ex. CC-36 (Product Images) at 13 (showing, coupled to Netlist DDR5 MRDIMM's PCB, SDRAMs, MDBs, and MRCD).

See also Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1 (advertising and depicting Netlist's DDR5 DIMMs as including PMIC, temperature sensor devices, SMBus, I2C, I3C, SPD, RCD, DBs, and DRAMs):



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DIMM Changes from DDR4 to DDR5

Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data+ 8 ECC 2 channel per DIMM
DfE Support	DB Only	RCD, DB, and DRAM

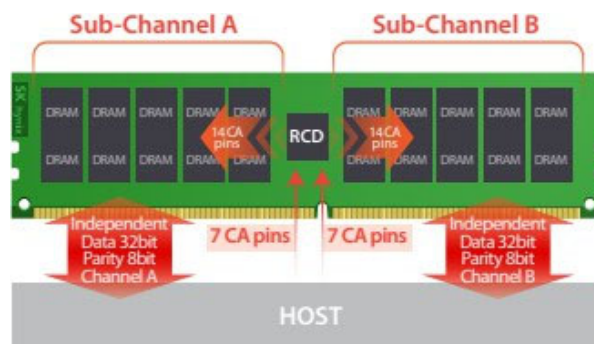
See also Ex. CC-86 (Netlist DDR5 Overclock RDIMM) at 1–2 (advertising Netlist’s DDR5 Overclock RDIMM as including features of DIMM Clock Driver, PMIC on DIMM, Temperature sensor device, SMBus, I2C, I3C, SPD Hub, RCD, and DRAMs:

FEATURES

- Data Rate (Mbps) : 5600~6800
- Voltage (VDDQ) : Upto 1.4
- DIMM Clock Driver : @8000 Mbps
- Power Supply : PMIC on DIMM
- Temp sensor device : 2 or 3 per DIMM
- Side bus protocol : Legacy SMBus, I2C, I3C
- Sideband bus access : IxC Hub in SPD
- Channel Architecture : 40-bit data channel, 32 data+ 8 ECC 2 channel per DIMM
- DfE Support : RCD and DRAM
- PCB Mechanical Outline : JEDEC (Height 31.25mm)
- Environmental : 0°C – 95°C, RoHS Compliant and Halogen-Free

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See also Ex. CC-40 (SK hynix A New Standard) at 5:

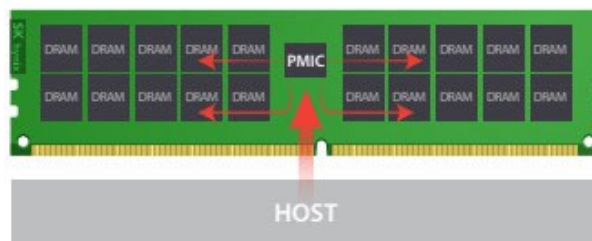
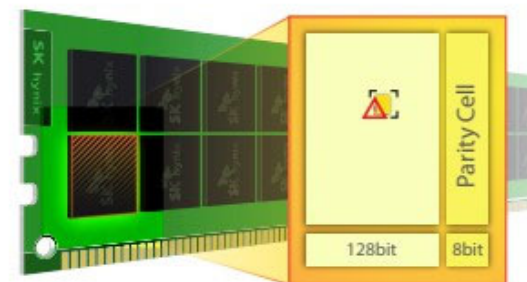


New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Enhanced Reliability

In-DRAM ECC on the DIMM offloads the system's error correction burden and considerably improves Reliability, Availability, and Serviceability (RAS).



Stable Power Management

On-board power management IC (PMIC) eliminates the need for external voltage regulation and ensures more stable power provision to the DRAM.

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[1.c.1] the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines,	Each of the set of components coupled to the PCB includes a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, <i>e.g.</i>, the DDR5 SDRAMs coupled to the C/A and data lines. For example, as shown in Netlist's DDR5 VLP RDIMM Datasheet, [REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 1. T [REDACTED] [REDACTED] See, <i>e.g.</i>, Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 19 (annotated): [REDACTED] [REDACTED] See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7, 8. [REDACTED]
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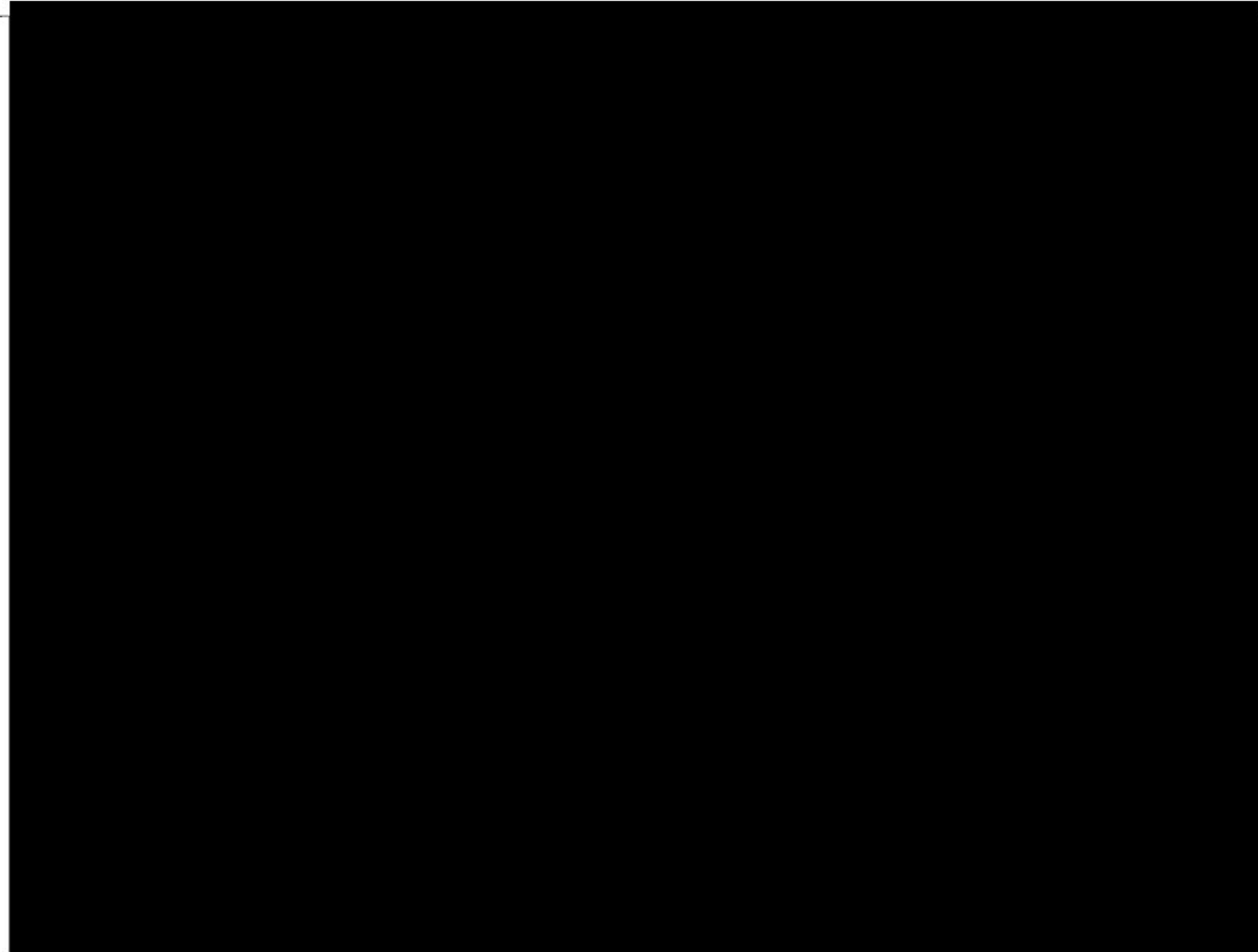
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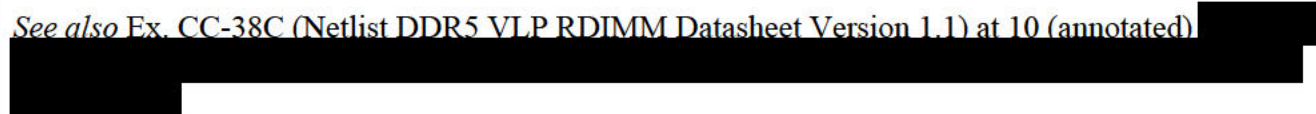
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 17, 19, 22, 24.

See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 10 (annotated)



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As another example, as shown in Netlist's DDR5 Overclock RDIMM Datasheet, [REDACTED]
[REDACTED] Ex. CC-39C (Netlist DDR5 Overclock
RDIMM Datasheet Version 1.0) at 1. [REDACTED]
[REDACTED] See, e.g., Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet
Version 1.0) at 19 (annotated):



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	<i>See also</i> Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–6. <i>See also</i> Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 11–12 (annotated)
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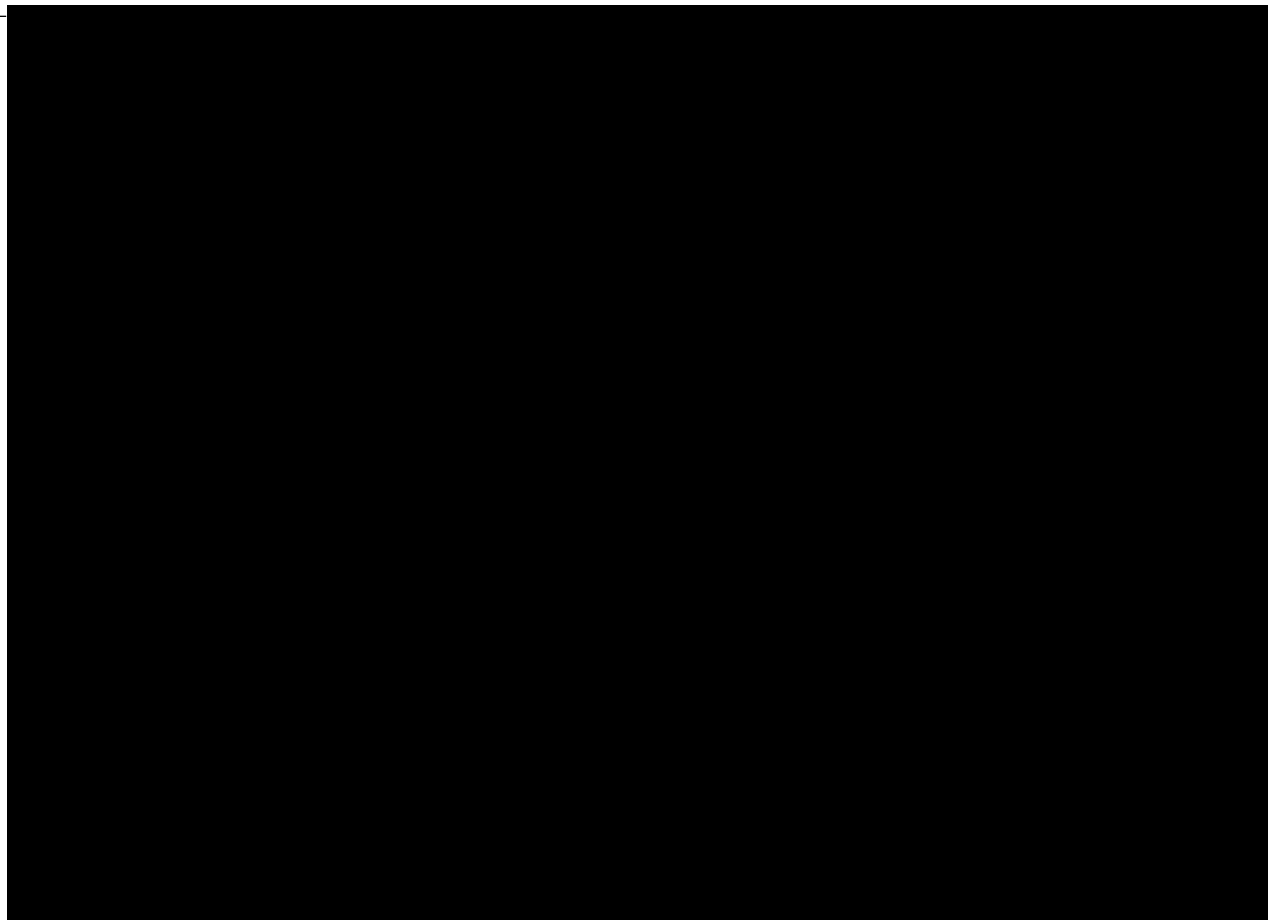


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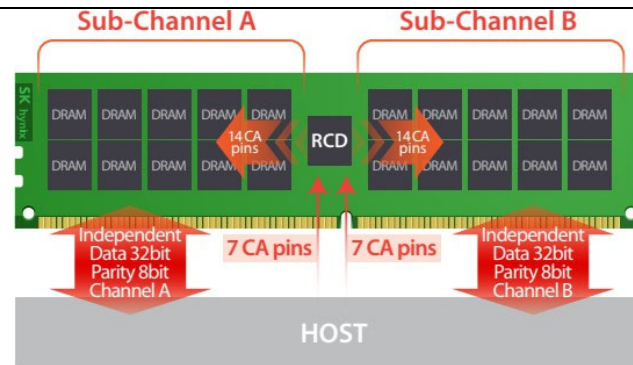
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Furthermore, Netlist advertises Netlist's VLP RDIMMs as including 10 or 20 DRAMs, based on the specific configuration. *See* Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 2 (showing number of DRAMs for Netlist's DDR5 VLP RDIMMs):

And, SK hynix depicts SK hynix DDR5 RDIMMs as including 20 DRAMs. Ex. CC-40 (SK hynix A New Standard) at 5.

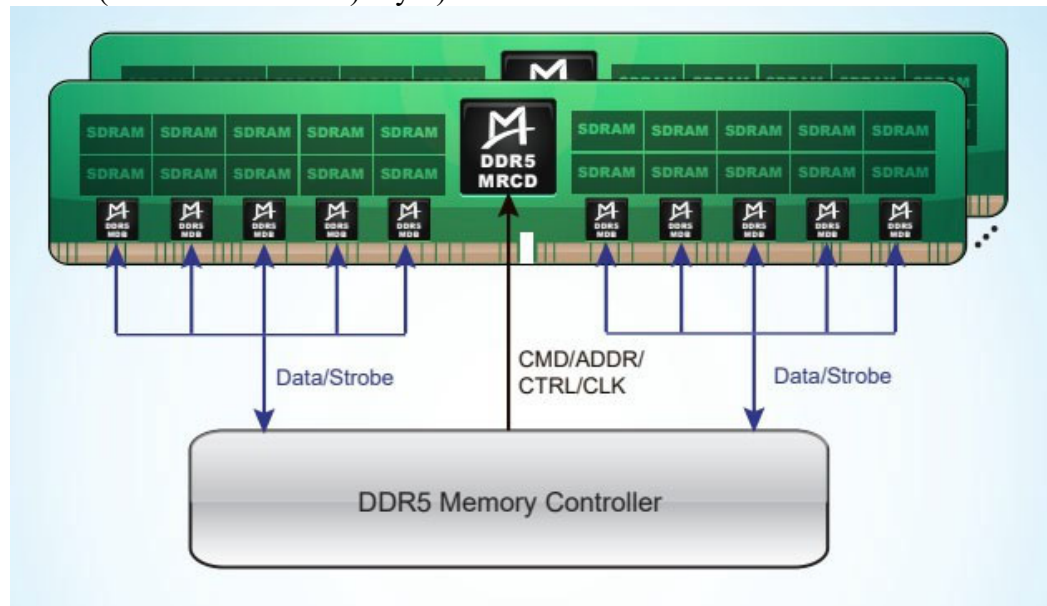
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New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Furthermore, Montage, a supplier of DDR5 MRCDs for Netlist's DDR5 MRDIMMs, advertises in Montage's DDR5 MRCD Flyer and Montage's DDR5 MDB flyer that Montage's MRCD receives C/A inputs and re-generates and propagates signals to the DRAMs in operation with Montage's DDR5 MDBs, which facilitate data transfer between the DRAMs and the DDR5 memory controller (e.g., host server or host CPU). *See, e.g.,* Ex. CC-12 (M88MR5RCD01 Flyer English) at 1–2; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1.





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	Additionally, Montage advertises DQ signal multiplexing via Montage's DDR5 MDBs. <i>See</i> Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1.
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[REDACTED]

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[1.c.2] the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

In each of the Practicing '366 Products, the DDR5 SDRAM devices are operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines.

For example, as illustrated in Neflist DDR5 VLP RDIMM Datasheet, [REDACTED]

[REDACTED]

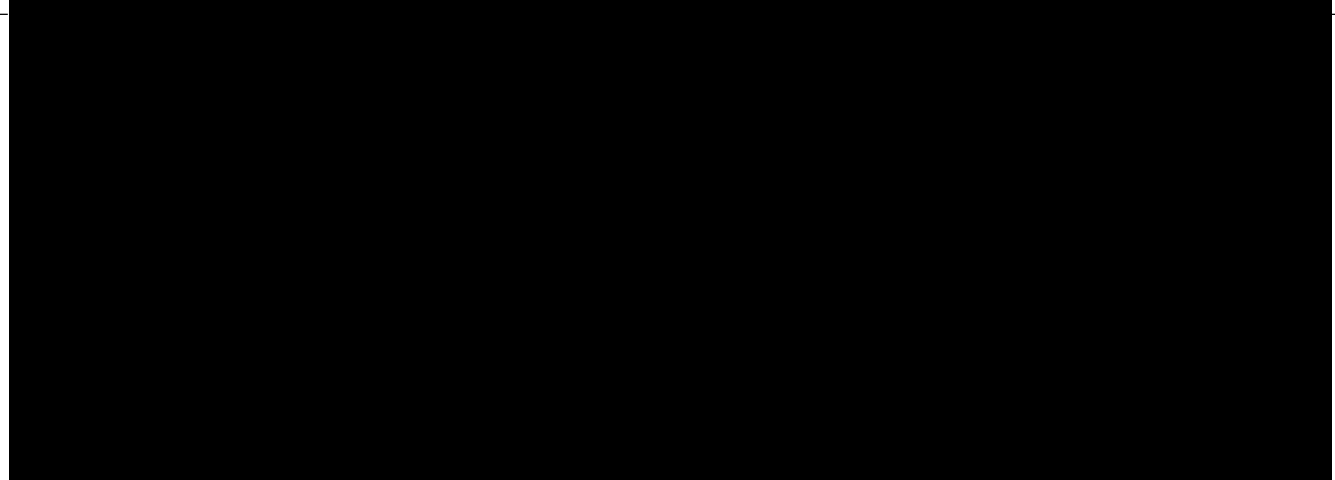
[REDACTED]



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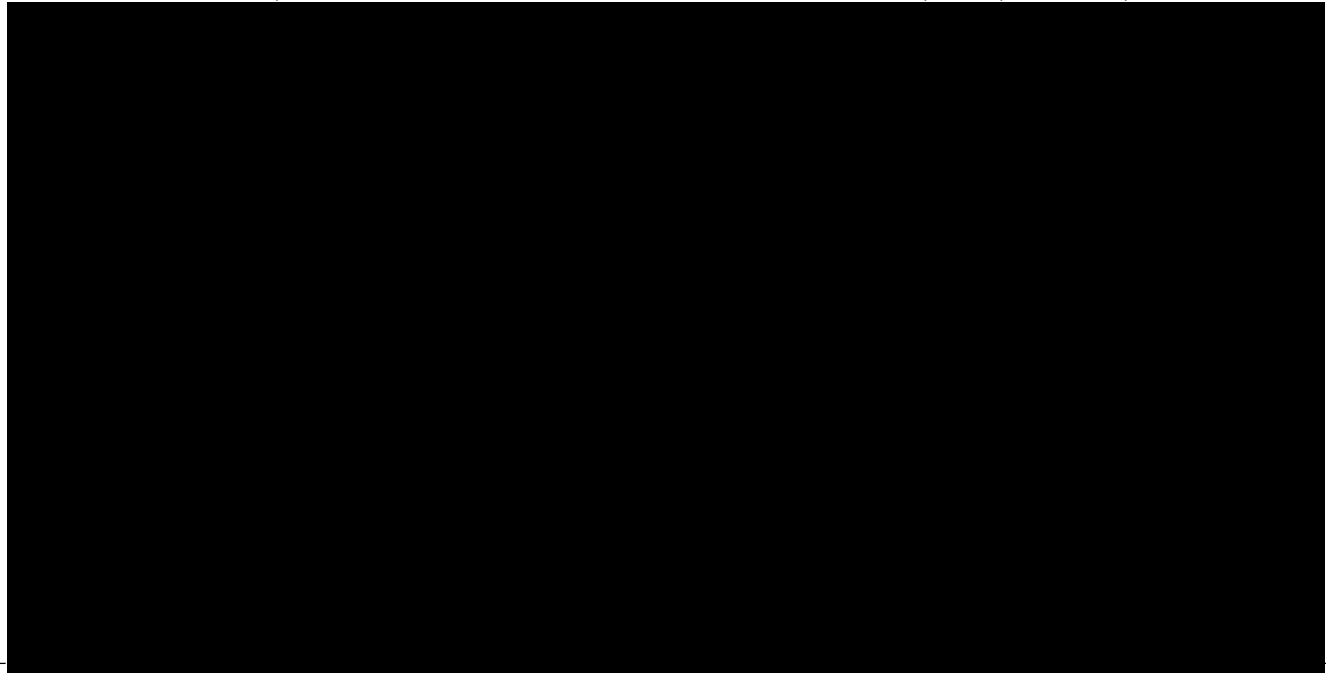
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 17, 19, 22, 24.

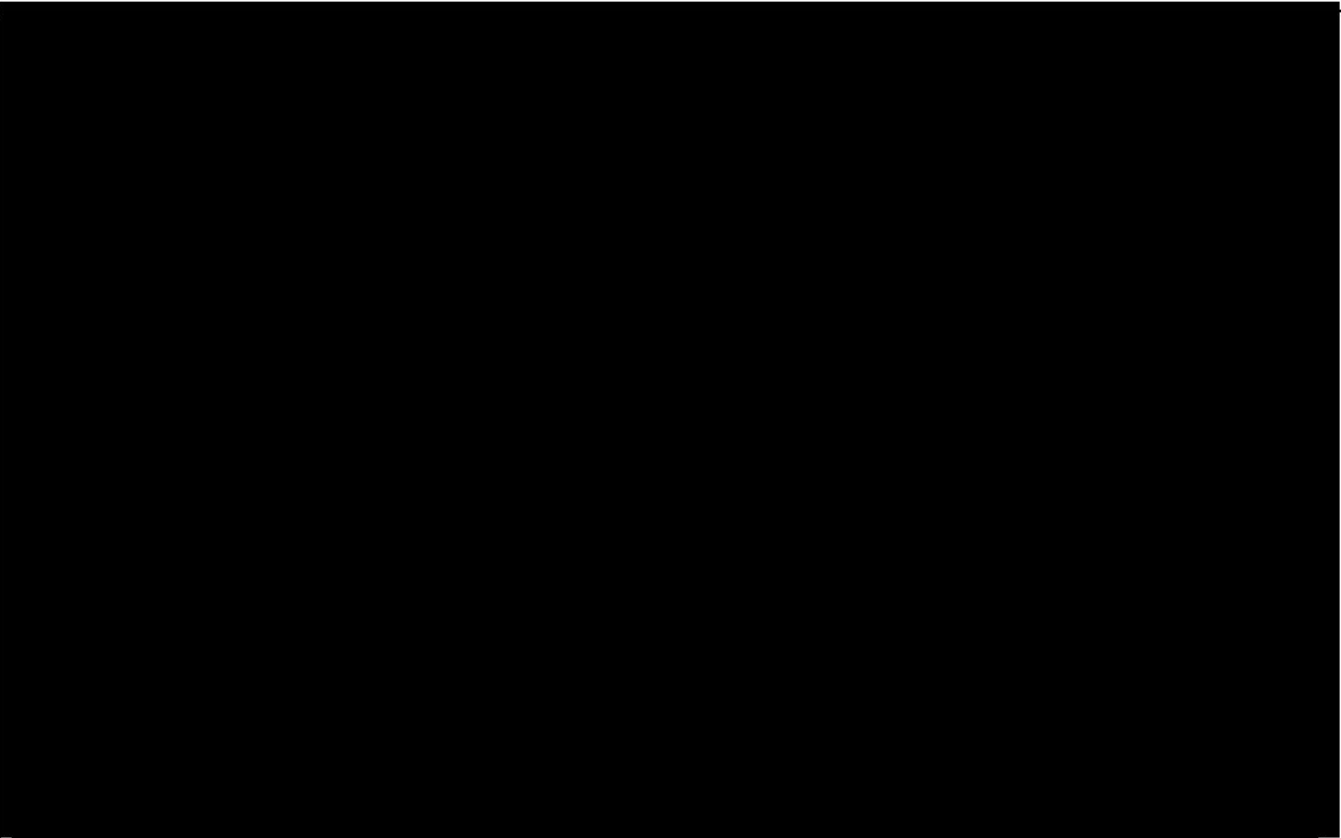
See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 7 (annotated):



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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9 (annotated):). See Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9 (annotated):
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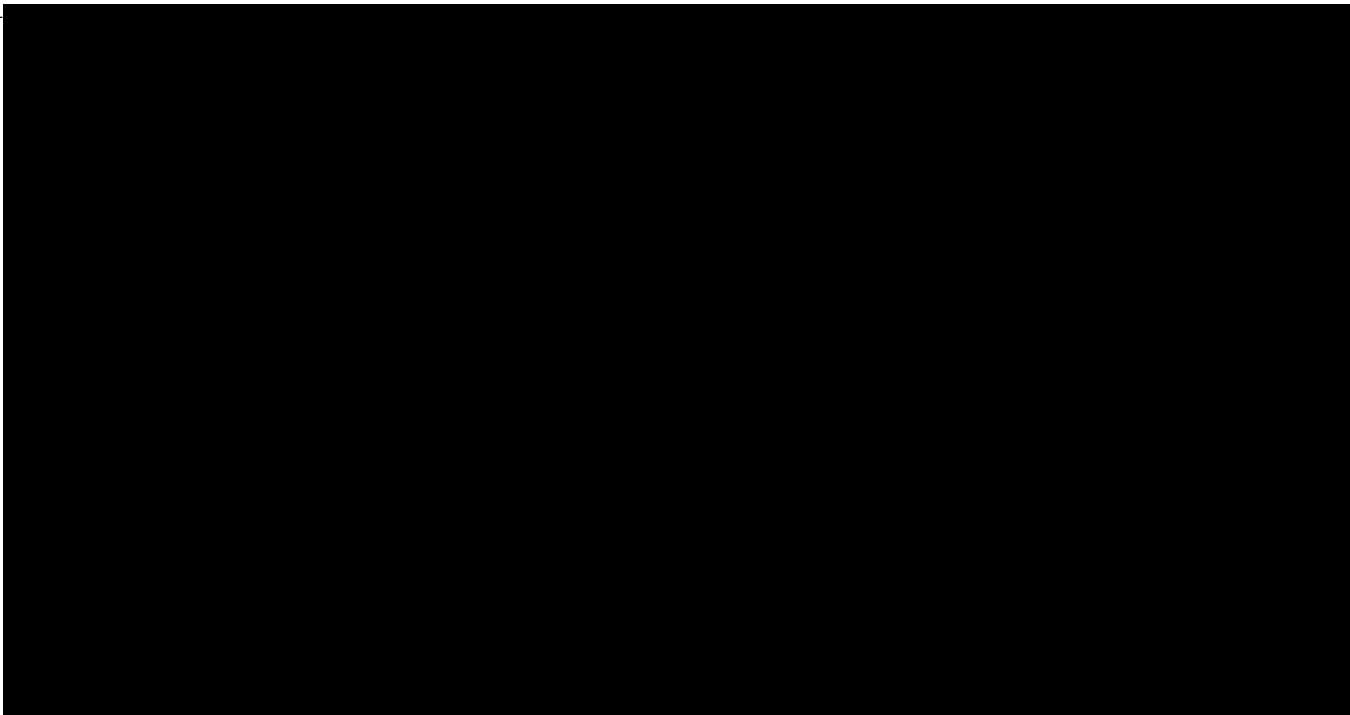
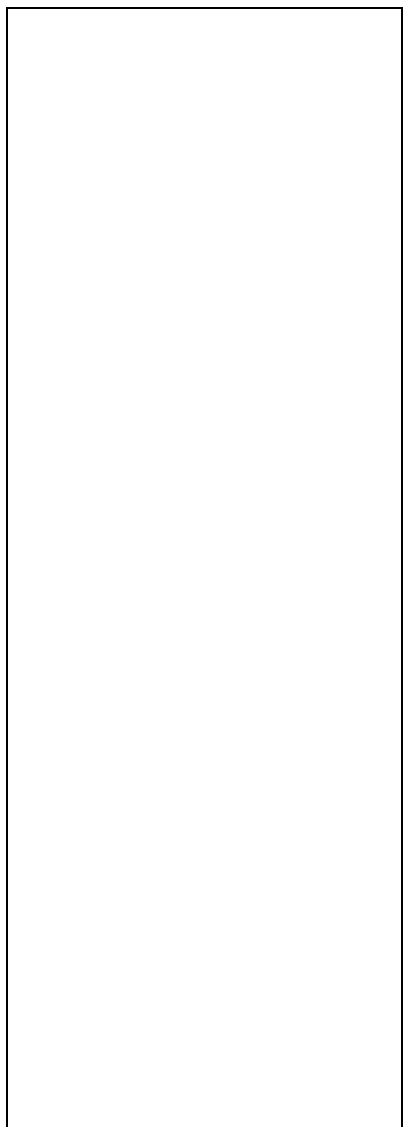


See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7, 8.

See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 7 (annotated):



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See also, e.g., Ex. CC-05 (JESD79-5C.01 v1.31) at 151 (DDR5 SDRAM command truth table, showing that write or read operations which result in receiving or outputting data along the set of data lines are in response to combination of C/A signal):

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4 DDR5 SDRAM Command Description and Operation

4.1 Command Truth Table

- (a) Notes 1, 2, and 14 apply to the entire Command truth table
- (b) To improve command decode time, the table has been optimized to orient all 1-cycle commands together and all 2-cycle commands together, allowing CA1 to be used to identify the difference between a 1-cycle and a 2-cycle command.

[BG=Bank Group Address, BA=Bank Address, R=Row Address, C=Column Address, MRA=Mode Register Address, OP=Op Code, CID=Chip ID, CW=Control Word, X=Don't Care, V=Valid].

Table 30 — Command Truth Table

Function	Abbreviation	CS_n	CA Pins														NOTES
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 17, 20
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 11, 13, 20
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 13, 21, 20
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	WR Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	V	V	CID3	

As another example,

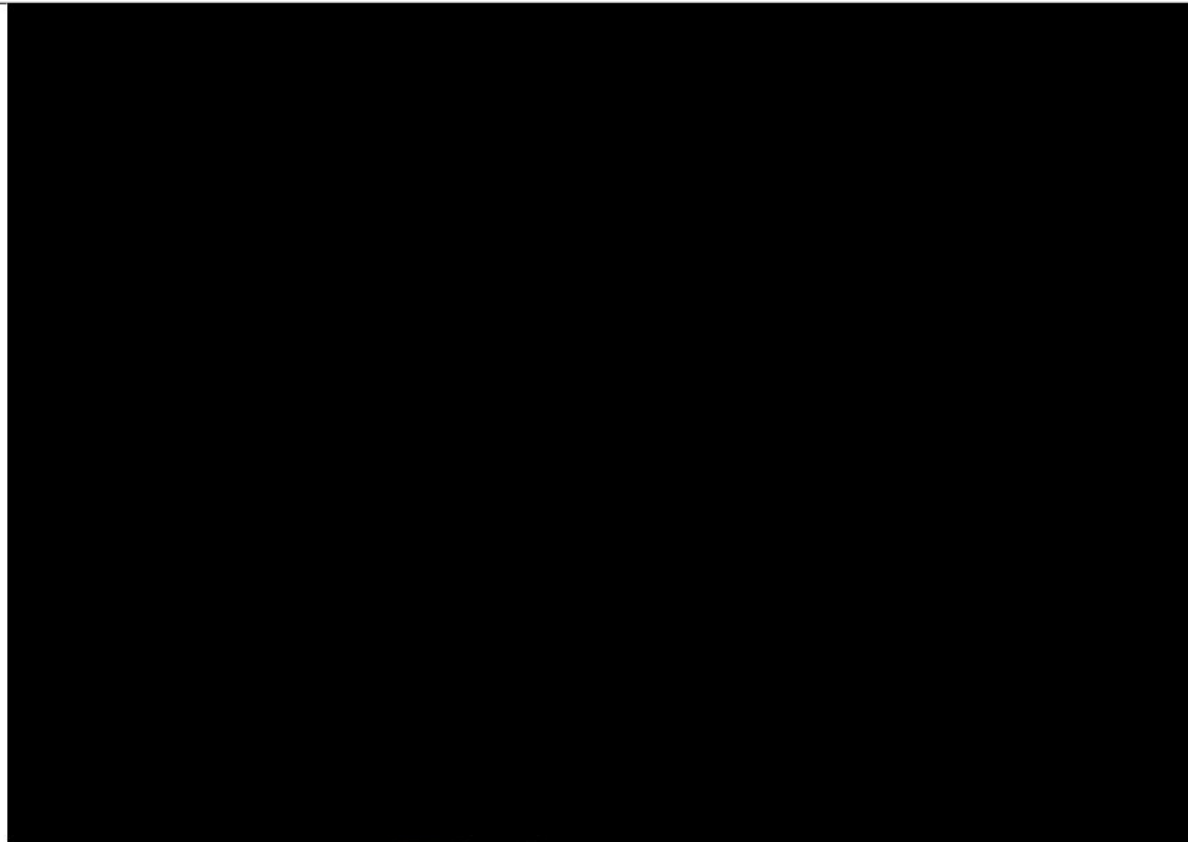
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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 11–12 (annotated).

The input and output of the data by SDRAMs are in response to C/A signals input to the SDRAMs. For example, as illustrated below, one input to the SDRAM are chip select (CS) signals. Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 11–12 (annotated):

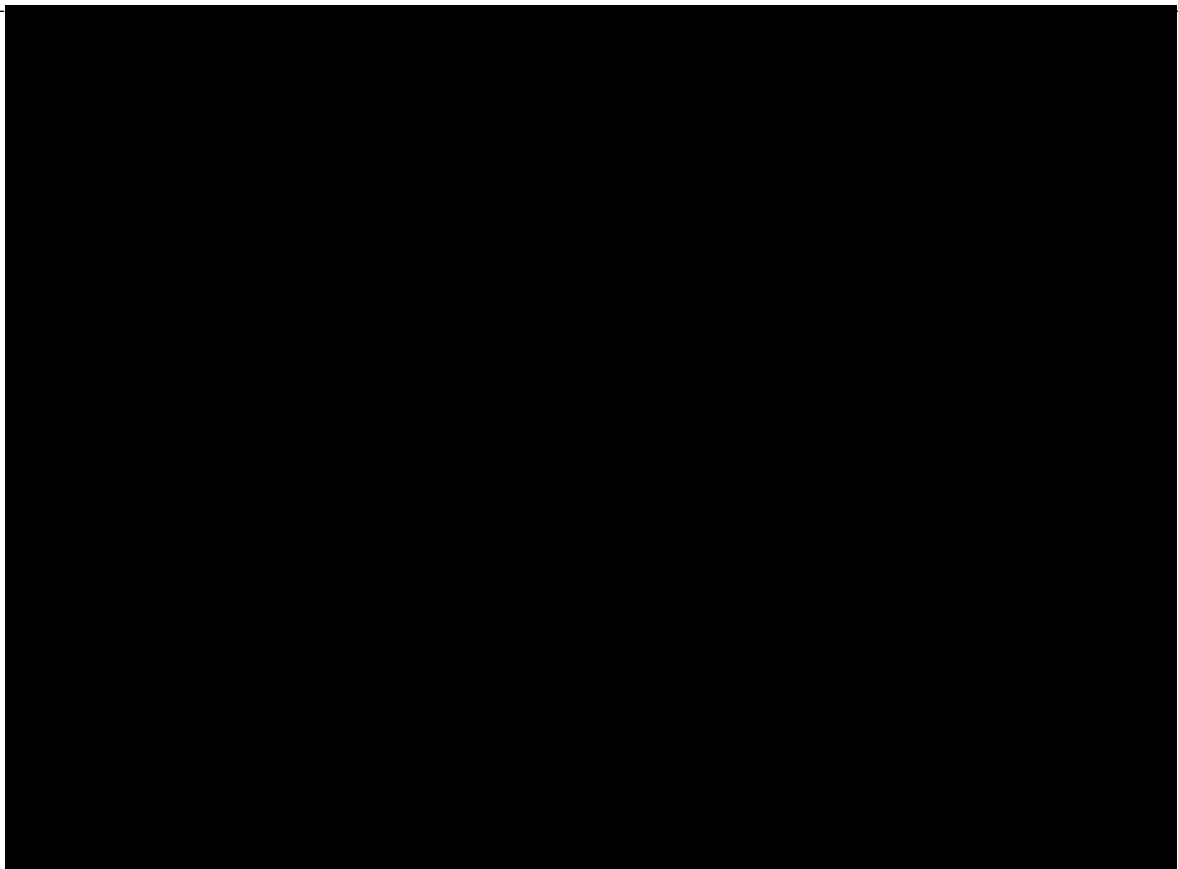
[Redacted]

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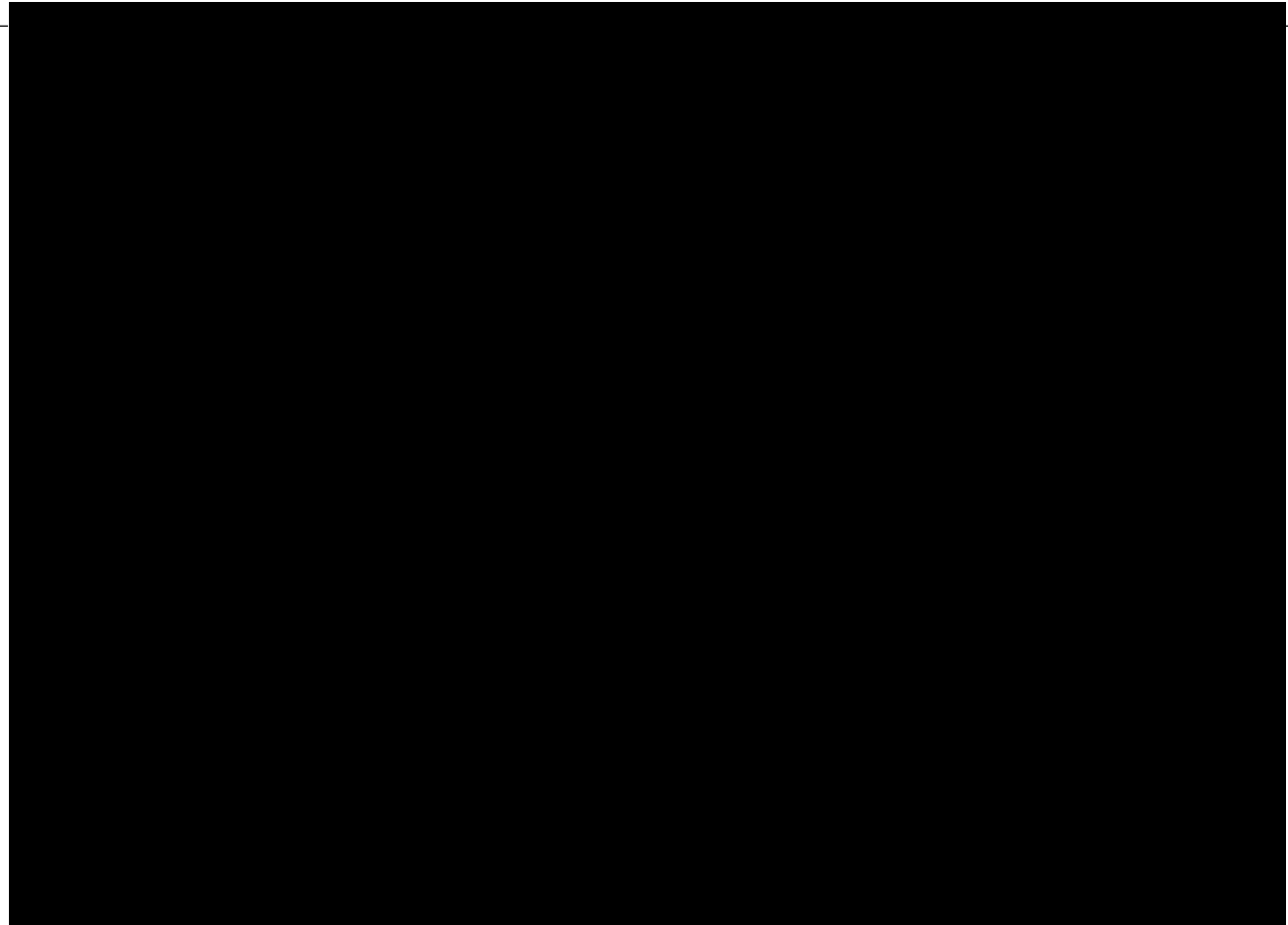


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See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10 (annotated):

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 See Ex. CC-39C
(Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 9 (annotated):

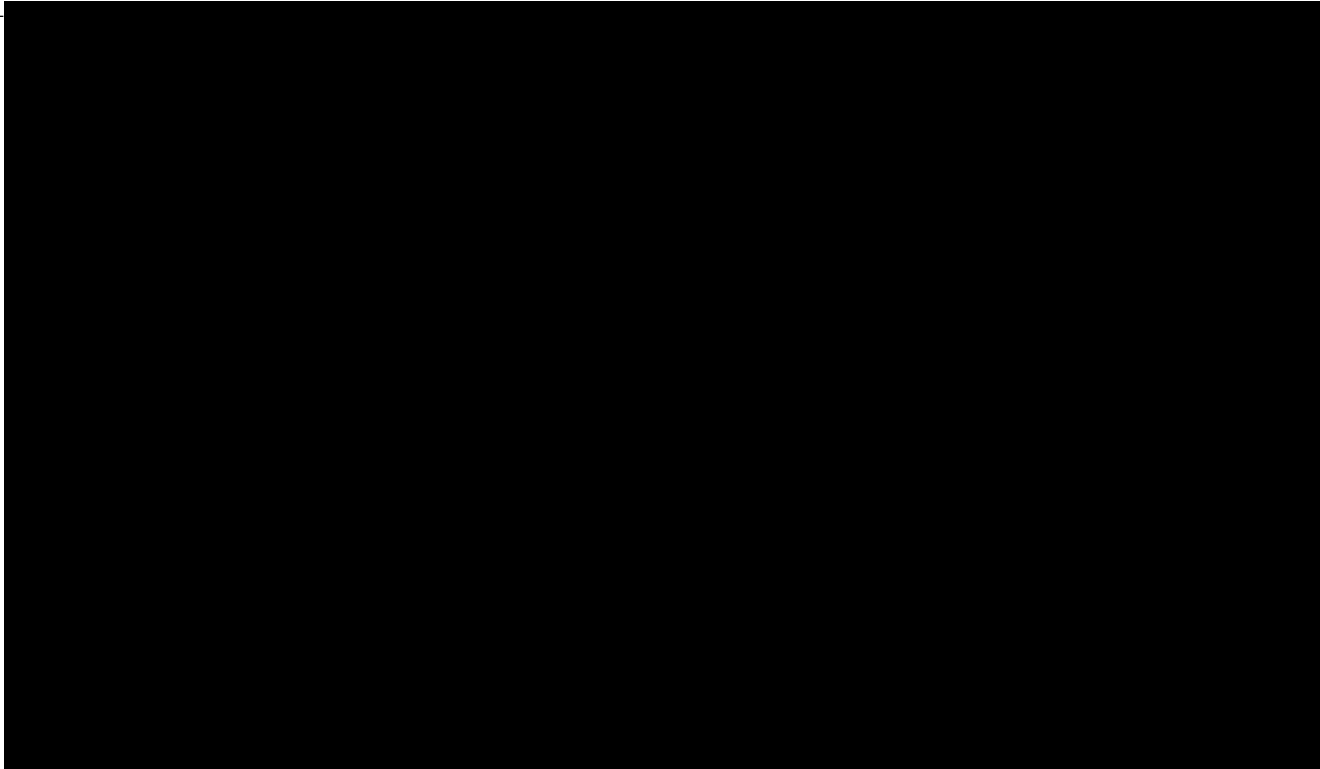
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see also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 5:



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See also, e.g., Ex. CC-05 (JESD79-5C.01 v1.31) at 151 (DDR5 SDRAM command truth table, showing that write or read operations which result in receiving or outputting data along the set of data lines are in response to combination of C/A signal):

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4 DDR5 SDRAM Command Description and Operation

4.1 Command Truth Table

- (a) Notes 1, 2, and 14 apply to the entire Command truth table
 (b) To improve command decode time, the table has been optimized to orient all 1-cycle commands together and all 2-cycle commands together; allowing CA1 to be used to identify the difference between a 1-cycle and a 2-cycle command.

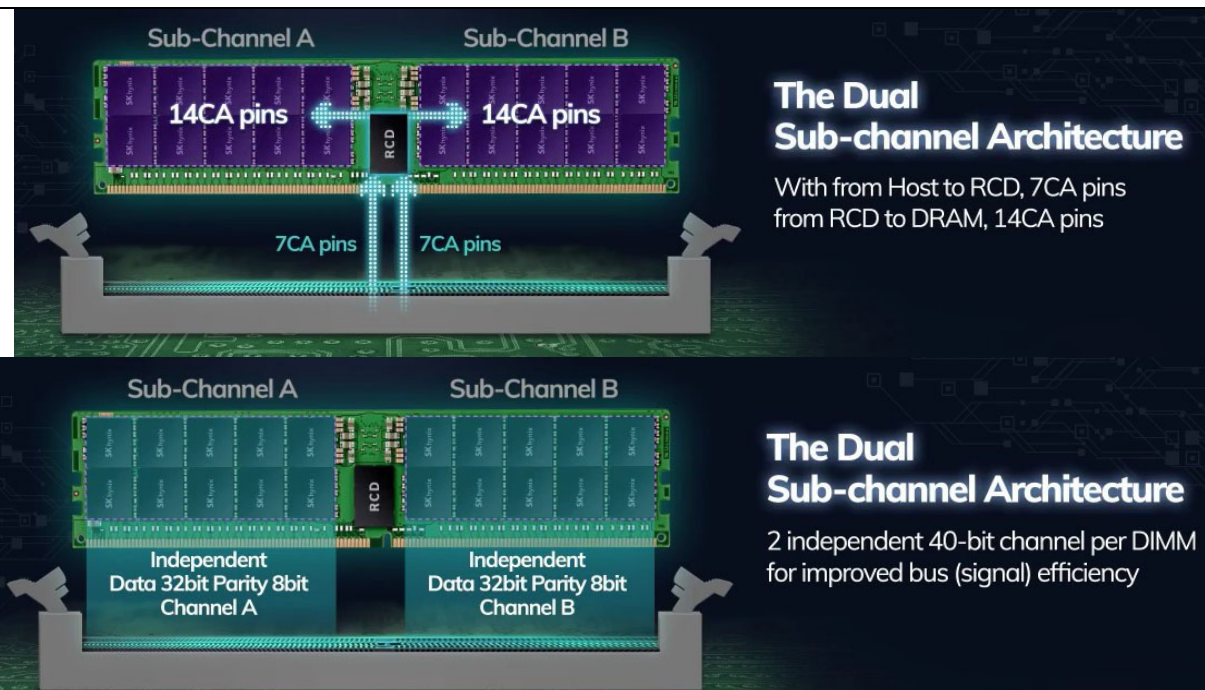
[BG=Bank Group Address, BA=Bank Address, R=Row Address, C=Column Address, MRA=Mode Register Address, OP=Op Code, CID=Chip ID, CW=Control Word, X=Don't Care, V=Valid].

Table 30 — Command Truth Table

Function	Abbreviation	CS_n	CA Pins													NOTES	
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12		CA13
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 17, 20
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/ CID3/ R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 11, 13, 20
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 13, 21, 20
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	WR Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	V	V	CID3	

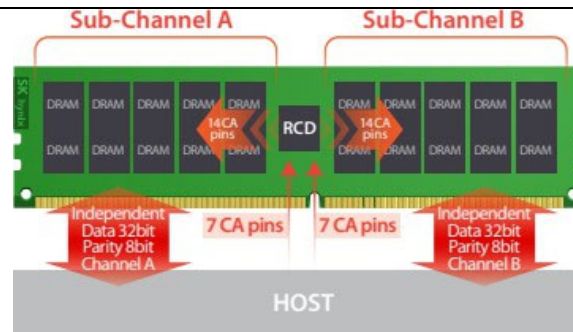
For another example, SK hynix advertises that SK hynix DDR5 RDIMMs propagate CA signals from the host to the plurality of SDRAMs to control data transfer between the two channels (Channel A and Channel B) and the host. *See* Ex. CC-41 (RDIMM SK hynix) at 4–5:

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See also Ex. CC-40 (SK hynix A New Standard) at 5:

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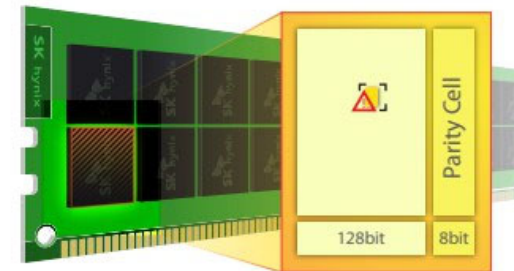


| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

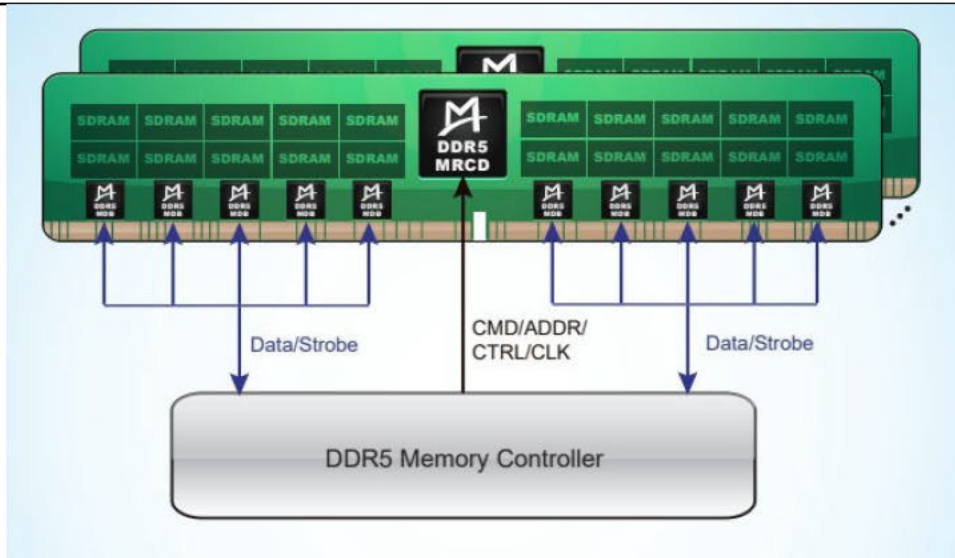
| Enhanced Reliability

In-DRAM ECC on the DIMM offloads the system's error correction burden and considerably improves Reliability, Availability, and Serviceability (RAS).



Furthermore, as discussed above in 1.c.1, Montage, a supplier of DDR5 MRCDs for Netlist's DDR5 MRDIMMs, advertises in Montage's DDR5 MRCD Flyer and Montage's DDR5 MDB flyer that Montage's MRCD receives C/A inputs and re-generates and propagates signals to the DRAMs in operation with Montage's DDR5 MDBs, which facilitate data transfer between the DRAMs and the DDR5 memory controller (e.g., host server or host CPU). *See, e.g.,* Ex. CC-12 (M88MR5RCD01 Flyer English) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1.

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Additionally, Montage advertises DQ signal multiplexing via Montage's DDR5 MDBs. *See* Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1.

Furthermore,



Ex. CC-36 (Product Images) at 9 (showing SK hynix DRAM on Netlist DDR5 RDIMM).

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Ex. CC-36 (Product Images) at 6 (showing SK hynix DRAM on Netlist DDR5 Overclock RDIMM).



Ex. CC-36 (Product Images) at 13 (showing SK hynix DRAM on Netlist DDR5 MRDIMM).

See also Ex. CC-53C (0887-99A-1M6060 Prelim 02-08-22) at 3

x. CC-54C (0888-97A-1A6B61-X00) at 2

And, SK hynix advertises on SK hynix's website that SK hynix DDR5 DRAMs are JEDEC-compliant.
Ex. CC-72 (SK hynix DDR5) at 1:

Next-generation Memory

Having developed the industry's first JEDEC-compliant DDR5 DRAM, SK hynix expects to lead the era of next-generation memory with its 16Gb or 24Gb DDR5. The performance enhancements and TCO savings over DDR4 realize greater value for customers, while the 20% smaller power footprint delivers value for the environment and society as well.

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Accordingly, based at least on the above information and the JEDEC standard for DDR5 DRAMs, the DDR5 DRAMs included on each of the Practicing '366 Products are operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines. *See, e.g.,* Ex. CC-05 (JESD79-5C.01 v1.31) at 151 (DDR5 SDRAM command truth table, showing that write or read operations which result in receiving or outputting data along the set of data lines are in response to combination of C/A signal):

4 DDR5 SDRAM Command Description and Operation

4.1 Command Truth Table

- (a) Notes 1, 2, and 14 apply to the entire Command truth table
 (b) To improve command decode time, the table has been optimized to orient all 1-cycle commands together and all 2-cycle commands together; allowing CA1 to be used to identify the difference between a 1-cycle and a 2-cycle command.

[BG=Bank Group Address, BA=Bank Address, R=Row Address, C=Column Address, MRA=Mode Register Address, OP=Op Code, CID=Chip ID, CW=Control Word, X=Don't Care, V=Valid].

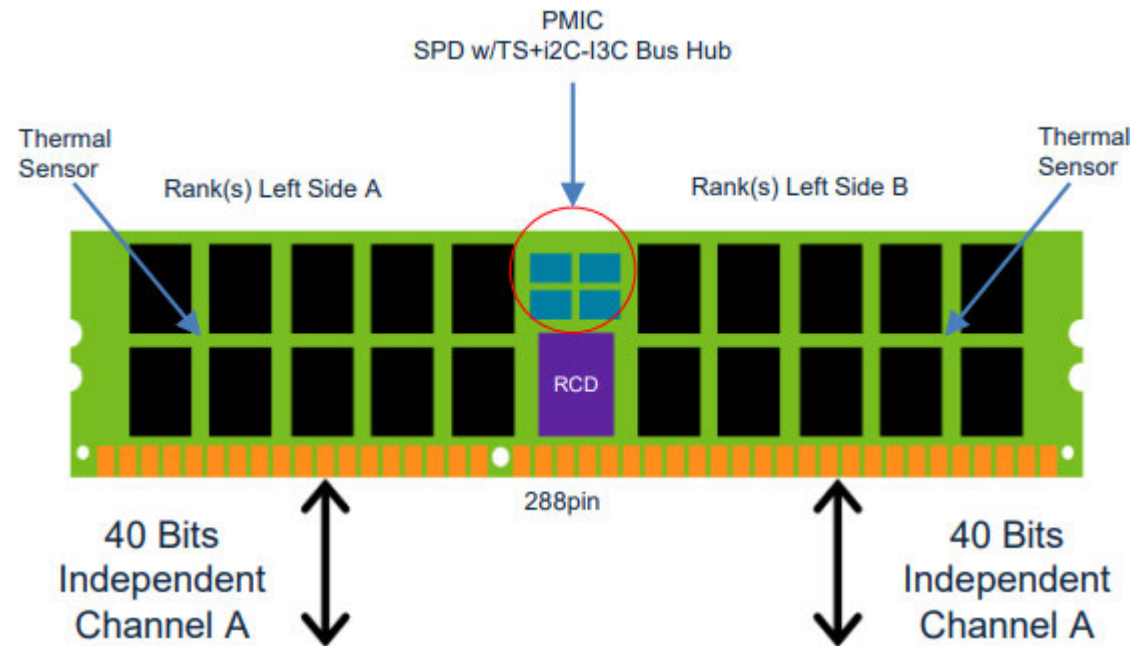
Table 30 — Command Truth Table

Function	Abbreviation	CS_n	CA Pins													NOTES	
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12		CA13
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 17, 20
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 11, 13, 20
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 13, 21, 20
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	VR Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	VR Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	V	V	CID3	

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[1.d] first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

In each of the Practicing '366 Products, there are at least four converter circuits contained in a PMIC coupled to the PCB.



Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

Example PMICs included in the Practicing '366 Products are shown below.

See Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10:

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See also Ex. CC-53C (0887-99A-1M6060 Prelim 02-08-22) at 3

at 3

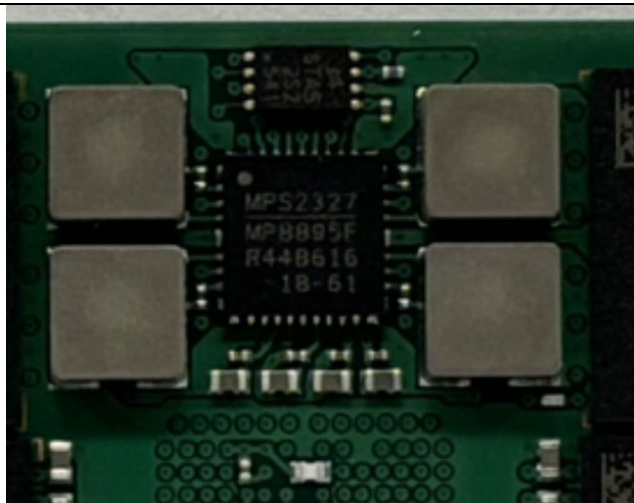
Ex. CC-54C (0888-97A-1A6B61-X00)

See also Ex. CC-78C (0887-06REVA-0114) at 4

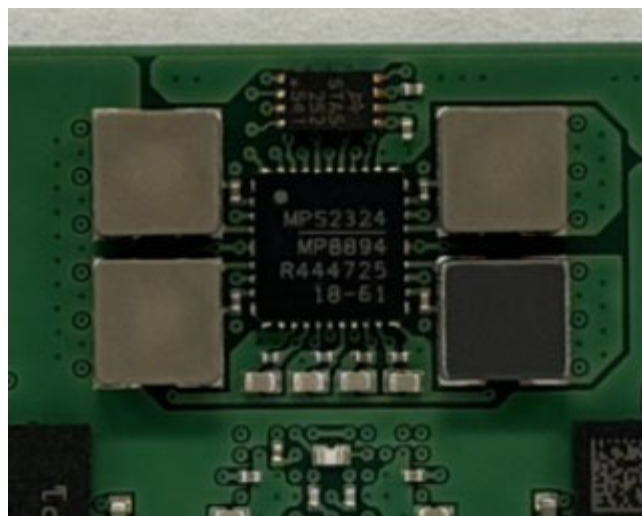
; Ex. CC-79C (0888-06REVA_0722) at 4

See also Ex. CC-36 (Product Images) at 3 (showing MPS MP8895F on Netlist DDR5 Overclock RDIMM):

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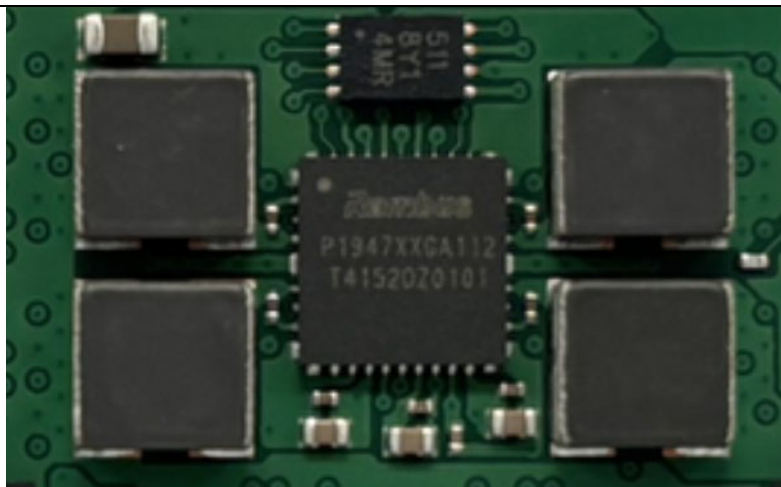


See also Ex. CC-36 (Product Images) at 10 (showing MPS MP8894 PMIC on Netlist DDR5 RDIMM):



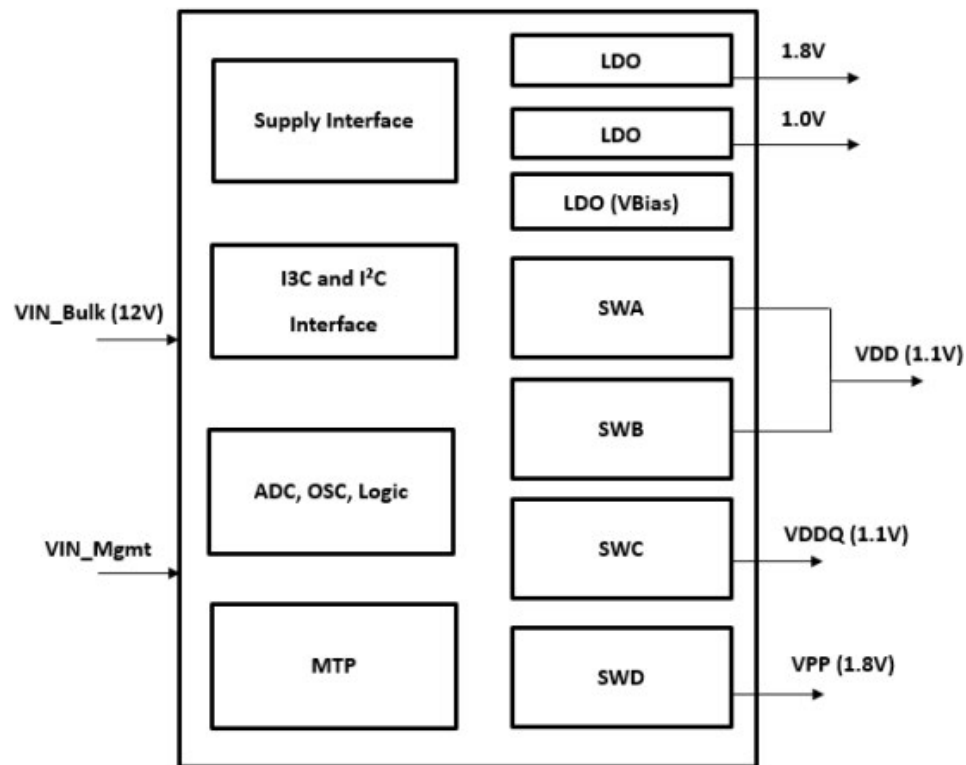
See also Ex. CC-36 (Product Images) at 16 (showing Montage P1947XXGA112 PMIC 5020 on Netlist DDR5 MRDIMM):

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart



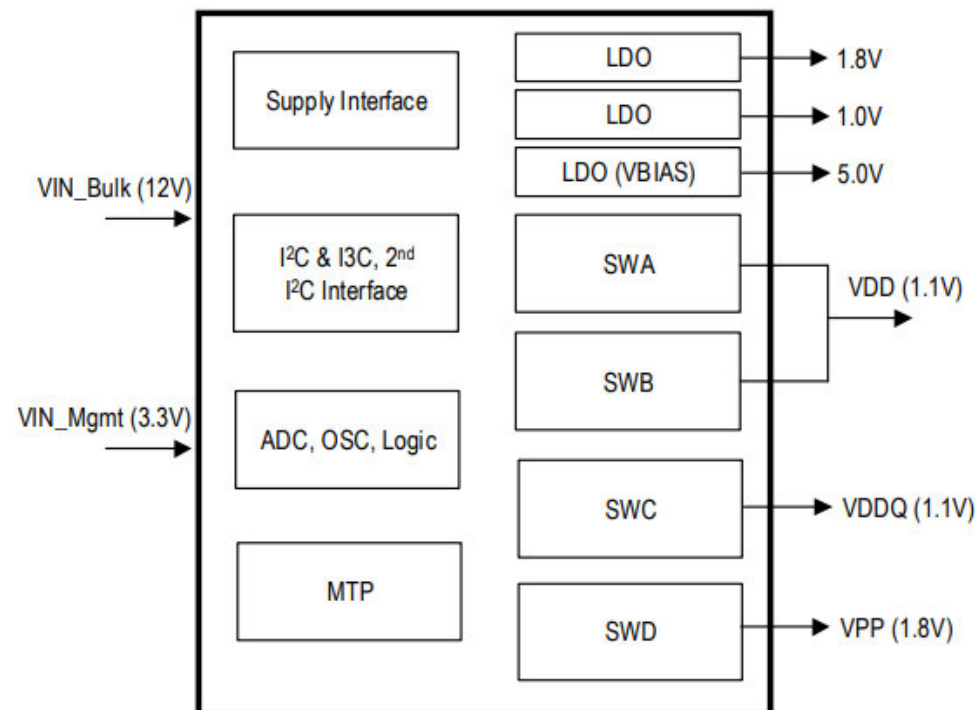
Each of the Practicing '366 Products includes a PMIC, which in turn includes at least four voltage converters, for example, step-down switching regulators and low-dropout (LDO) regulators. *See, e.g.*, Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1 (describing and showing that the Renesas P8900 PMIC, included in Netlist's 2Rx8 DDR5 VLP RDIMM, supports "[f]our step-down switching regulators: SWA, SWB, SWC, and SWD" and "[t]hree LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V"):

U.S. Patent No. 12,373,366 Domestic Industry Claim Chart

Block Diagram

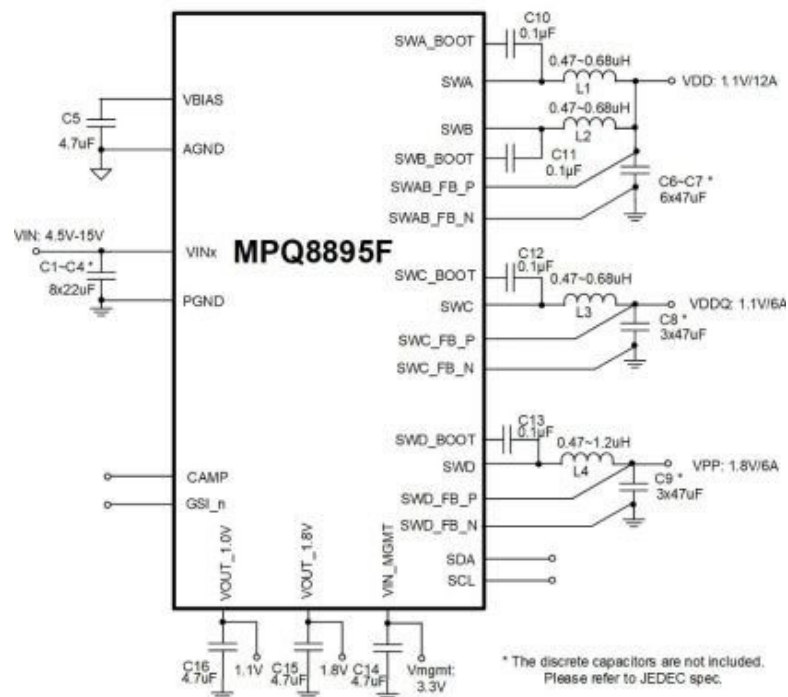
Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 (describing and showing that the Renesas P8910 PMIC, included in Netlist's 1Rx8 DDR5 VLP RDIMM, supports "[f]our step-down switching regulators: SWA, SWB, SWC, and SWD" and "[t]hree LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V"):

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Block Diagram

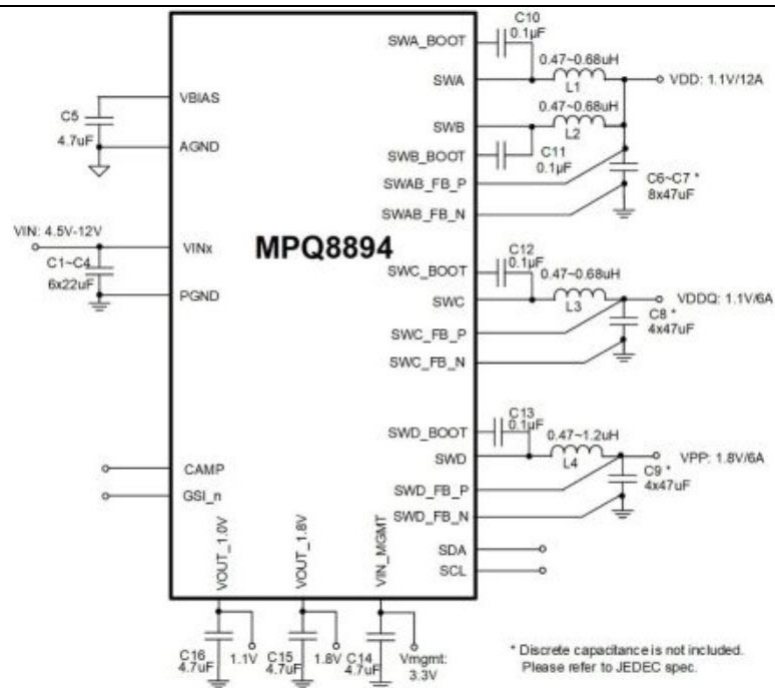
Ex. CC-28 (MPQ8895F) (describing and showing that the MPS MP8895F PMIC, included in Netlist's DDR5 Overclock RDIMM, "integrates four high-efficiency, step-down DC/DC converters," Buck A, B, C, and D, all of which are 6A DC/DC converters, and two low-dropout (LDO) regulators (VOUT_1V and VOUT_1.8V));

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Ex. CC-29 (MPQ8894) (describing and showing that the MPS MP8894 PMIC, included in Netlist's DDR5 RDIMM, "integrates four high-efficiency, step-down DC/DC converters," Buck A, B, C, and D, all of which are 4A DC/DC converter, and two low-dropout (LDO) regulators (VOUT_1V and VOUT_1.8V));

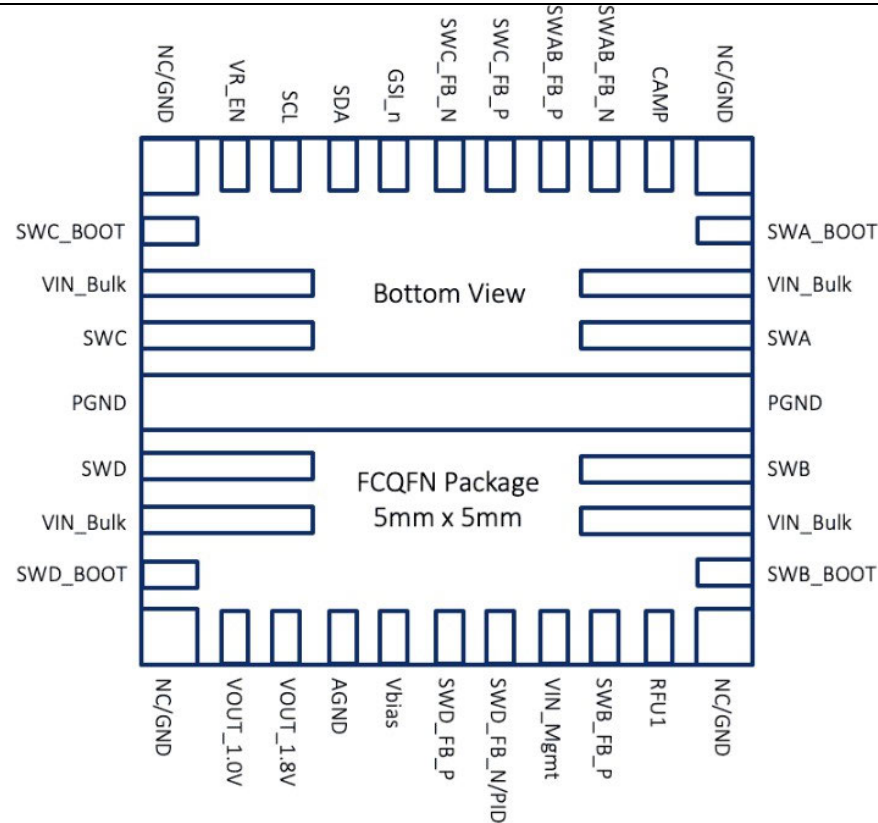
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Ex. CC-76 (Rambus DDR5 PMICs) at 2 (describing and showing that Rambus's PMIC 5020, included in Netlist's DDR5 MRDIMM, includes "Four step-down switching regulators (SWA, SWB, SWC, and SWD) and 3 LDO regulators (VBias, VOUT_1.8V, VOUT_1.0V)"):

[REDACTED]

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Additionally, as noted above in Claim 1[b], the PMICs used in Netlist's DDR5 RDIMMs comply with the appropriate JEDEC standards. *See* Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1 ("The P8900 is a power management IC (PMIC) designed for typical DDR5 RDIMM, DDR5 LRDIMM, and DDR5 NVDIMM applications."); Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1 ("The PMIC is fully compliant to the JEDEC DDR5 PMIC (PMIC5010; Low Current PMIC) specification."); Ex. CC-29 (MPQ8894) ("JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev. 1.8 Compliant"); Ex. CC-28 (MPQ8895F) ("JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev 1.8 Compliant"); Ex. CC-76 (Rambus DDR5 PMICs) at 1 ("Meets or exceeds all JESD300-5 performance specifications"); Ex. CC-80 (Rambus Expands Chipset) at 1 ("The Rambus DDR5 server PMIC family includes products for the JEDEC extreme current (PMIC5020) . . .

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specifications.”).

According to the JEDEC PMIC standards, for example, the PMIC 5000/5010 and PMIC 5020 standards, the PMICs included in the Practicing '366 Products include “[f]our step down switching regulators: SWA, SWB, SWC, and SWD” and “3 LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V.” See Ex. CC-81 (JESD301-1A) at 2 (annotated):

2.1.1 Common Features Summary

Table 1 — PMIC Device Type Summary

Device Type	SWA	SWB	SWC	SWD	Unit
PMIC5000 - Current Capability per Phase	5	5	5	5	A
PMIC5010 - Current Capability per Phase	3	3	3	3	A

- VIN_Bulk input supply range: 4.25 V to 15.0 V
- VIN_Mgmt input supply range: 3.0 V to 3.6 V
- Four step down switching regulators: SWA, SWB, SWC and SWD
- Programmable dual phase and single phase regulator for SWA and SWB
- 3 LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V
- Automatic switchover from VIN_Mgmt input supply to VIN_Bulk input supply
- Error injection capability
- Persistent Error log registers
- Write protect mode and programmable of operation
- Independently programmable output voltages, power up and power down sequence for switch regulators
- Input and output power good status reporting mechanism
- VIN_Bulk input supply protection feature: Input over voltage
- Output switch regulators protection feature: Output over voltage, output under voltage, output current limiter
- Output current and power measurement, output current threshold mechanism
- Temperature measurement, temperature warning threshold, critical temperature shutdown
- Multi Time Programmable Non-Volatile Memory
- Programmable and DIMM specific registers for customization
- General Status Interrupt Function
- Flexible Open Drain IO (I²C) and Push Pull (I³C Basic) IO Support

See also Ex. CC-83 (JESD301-4A) at 2 (annotated):

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1.2.2 Common Features summary

Table 1 — PMIC Device Type Summary

Device Type	SWA	SWB	SWC	SWD	Unit
PMIC5020 - Current Capability per Phase	8.5	8.5	10	3	A

- VIN_Bulk input supply range: 4.25 V to 15.0 V
- VIN_Mgmt input supply range: 3.0 V to 3.6 V
- Four step down switching regulators: SWA, SWB, SWC & SWD
- Programmable dual phase and single phase regulator for SWA and SWB
- 3 LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V
- Automatic switchover from VIN_Mgmt input supply to VIN_Bulk input supply
- Error injection capability
- Persistent Error log registers
- write-protect mode and programmable of operation
- Independently programmable output voltages, power up and power down sequence for switch regulators
- Input and output power good status reporting mechanism
- VIN_Bulk input supply protection feature: Input over voltage
- Output switch regulators protection feature: Output over voltage, output under voltage, output current limiter
- Output current and power measurement, output current threshold mechanism
- Temperature measurement, temperature warning threshold, critical temperature shutdown
- Multi Time Programmable Non-Volatile Memory
- Programmable and DIMM specific registers for customization
- General Status Interrupt Function
- Flexible Open Drain IO (I²C) and Push Pull (I3C Basic) IO Support

See, e.g., Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7–8
 CC-78C (0887-06REVA-0114) at 4 (

x. CC-79C (0888-06REVA_0722) at

4

Ex. CC-76 (Rambus DDR5 PMICs) at 2 (showing and describing VIN-Bulk

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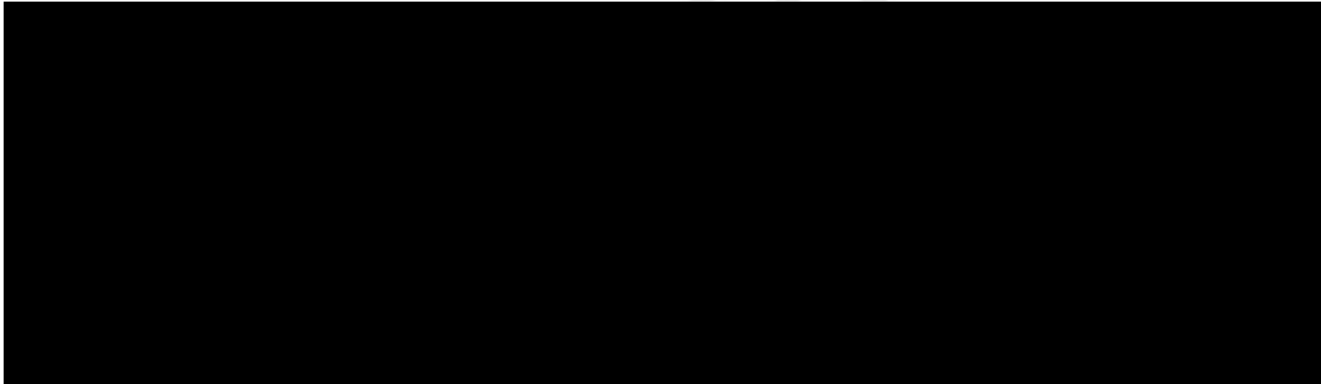
and VIN_Mgmt voltage inputs as well as various voltage outputs of the Rambus PMIC 5020). For instance, when VIN_MGMT is operating normally, VDD, VDDQ and VPP are generated from VIN_BULK while VDDIO and VDDSPD are generated from VIN_MGMT. However, when VIN_MGMT drops below a threshold, VDDIO and VDDSPD are generated from the 12V VIN_BULK as well. *See* Ex. CC-83 (JESD301-4A) at 21 (annotated):

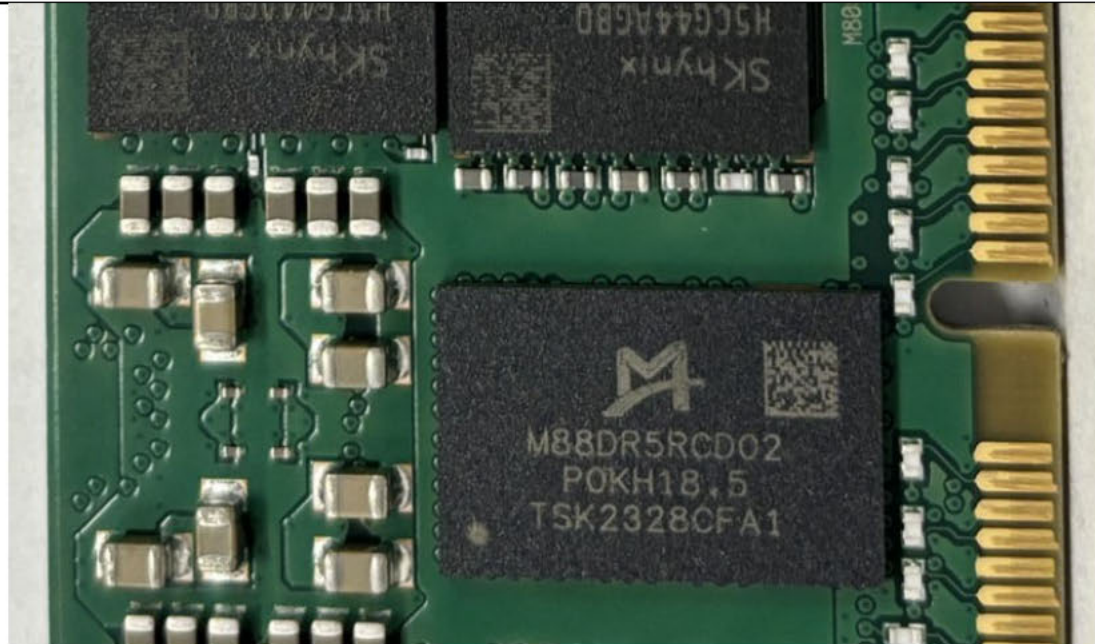
After host issues VR Enable command to the PMIC, the PMIC offers the input supply switchover function. The PMIC has an automatic internal input supply switchover function from VIN_Mgmt input supply to VIN_Bulk input supply. The PMIC triggers the switchover to VIN_Bulk input supply when VIN_Mgmt input supply drops below the threshold set in register Table 133, "Register 0x2F" [7]. The internal input supply switchover is for PMIC's

VOUT_1.8V & VOUT_1.0V LDO output. The PMIC's I²C/I³C Basic interfaces (SCL/SDA) are kept alive when PMIC switches over to VIN_Bulk input supply. Figure 13 shows automatic internal switchover function when VIN_Mgmt input supply drops below the threshold while maintaining its LDO outputs as well as I²C/I³C Basic interfaces. Under the switchover conditions VIN_Mgmt back-feed voltage shall be less than 0.2 V. While PMIC is in switchover mode to VIN_Bulk, the VIN_Mgmt input supply can re-power backup at any time and PMIC switches back to VIN_Mgmt input supply for its LDO outputs and I²C/I³C interface continues to operate as normal.

Additionally, as the various PMIC data sheets' block diagrams indicate, the four buck converters output VDD, VDDQ, VPP as specified and the at least two LDOs output respectively 1.0V and 1.8V output. *See* Ex. CC-75 (Renesas P8900 Short Form Datasheet) at 1; Ex. CC-74 (Renesas P8910 Short Form Datasheet) at 1; Ex. CC-28 (MPQ8895F); Ex. CC-29 (MPQ8894); Ex. CC-76 (Rambus DDR5 PMICs).

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[1.d.1] wherein the set of components includes: a first component configured to receive power via at least the first regulated voltage line; a second component configured to receive power via at least the second regulated voltage line; a third component configured to receive power via at least the third regulated voltage line; and a fourth component configured to receive power via at least the fourth regulated voltage line;	In each of the Practicing '366 Products, the set of components includes: • a first component configured to receive power via at least the first regulated voltage line; • a second component configured to receive power via at least the second regulated voltage line; • a third component configured to receive power via at least the third regulated voltage line; and • a fourth component configured to receive power via at least the fourth regulated voltage line. For example, as shown by Netlist's DDR5 Overclock RDIMM Datasheet at 13, any of SDRAMs D1-D40 are designed to receive VDD, VPP and VDDQ output by the buck converters; SPD_HUB is designed to receive VDDSPD and/or VDDIO, thermal sensors are shown as receiving VDDSPD and/or VDDIO, and RCD is designed to receive 1.0V VDDIO from 1.0V LDO and although not shown, also VDDQ(*).  Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 13. *As noted above, although not shown, RCD also receives a 1.1V voltage. This would be understood as for I/O communication with SDRAM for C/A signal transmissions. The 1.0V VDDIO is for operating the sideband bus (I2C/I3C). For example, Montage, one DDR5 RCD supplier, supplies the M88DR5RCD02 that is included in Netlist's DDR5 Overclock RDIMM. <i>See</i> Ex. CC-36 (Product Images) at 2 (showing M88DR5RCD02 included on Netlist's DDR5 Overclock RDIMM):
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[REDACTED]
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Furthermore, Montage states on its web site that its RCD receives “1.1V VDD and 1.0V VDDIO voltages.” *See* Ex. CC-87 (Montage M88DR5RCD02) at 1.

As an additional example, as shown by schematics of Netlist’s 1Rx8 DDR5 VLP RDIMM, [REDACTED]

[REDACTED] *See* Ex. CC-78C
(0887-06REVA-0114) at 2–4, 6 (annotated):



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As yet an additional example, as shown by schematics of Netlist's 2Rx8 DDR5 VLP RDIMM.

See Ex. CC-79C

(0888-06REVA_0722) at 2-4, 6 (annotated):

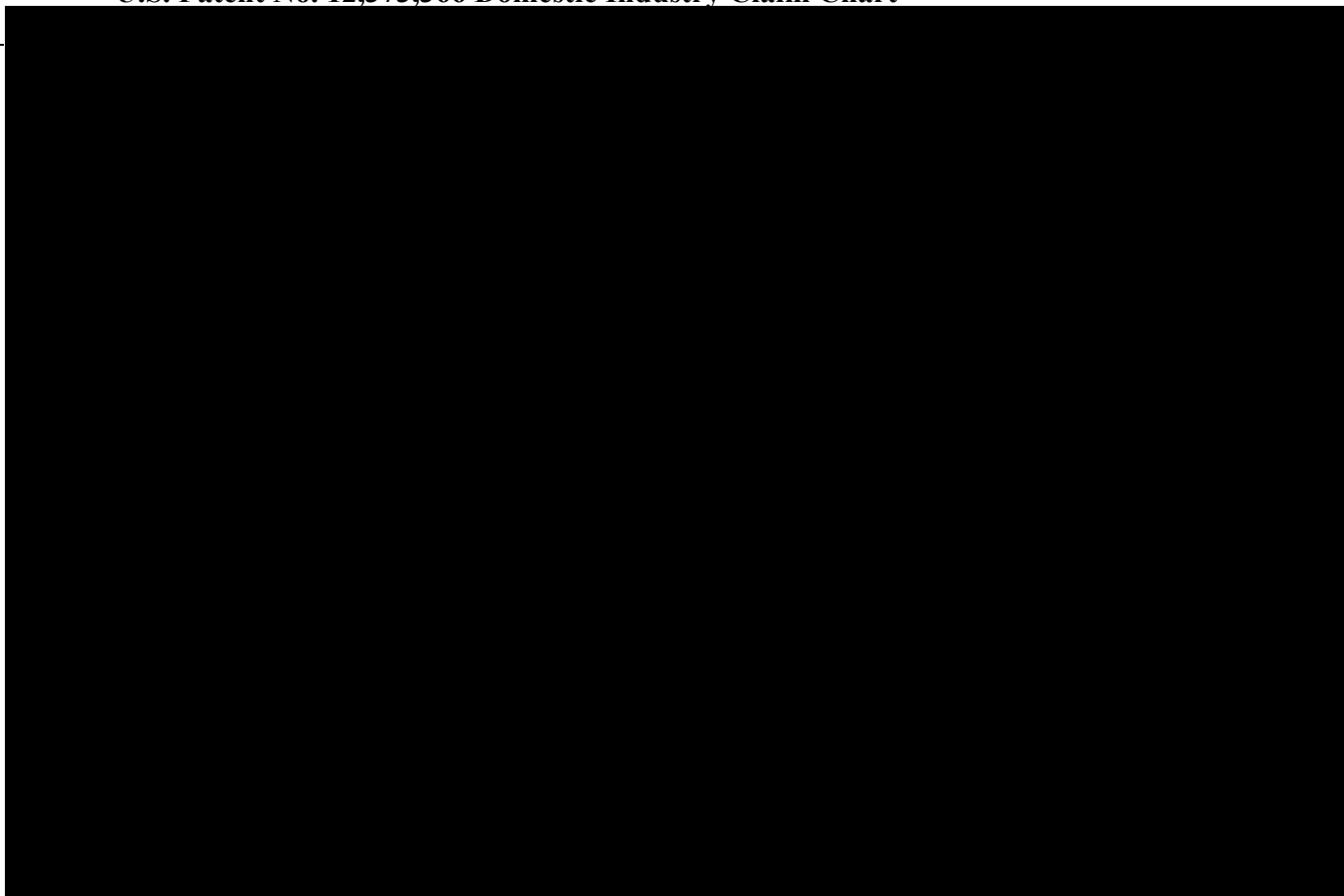


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	A large solid black rectangular area covering the majority of the table's content, indicating redacted information.		

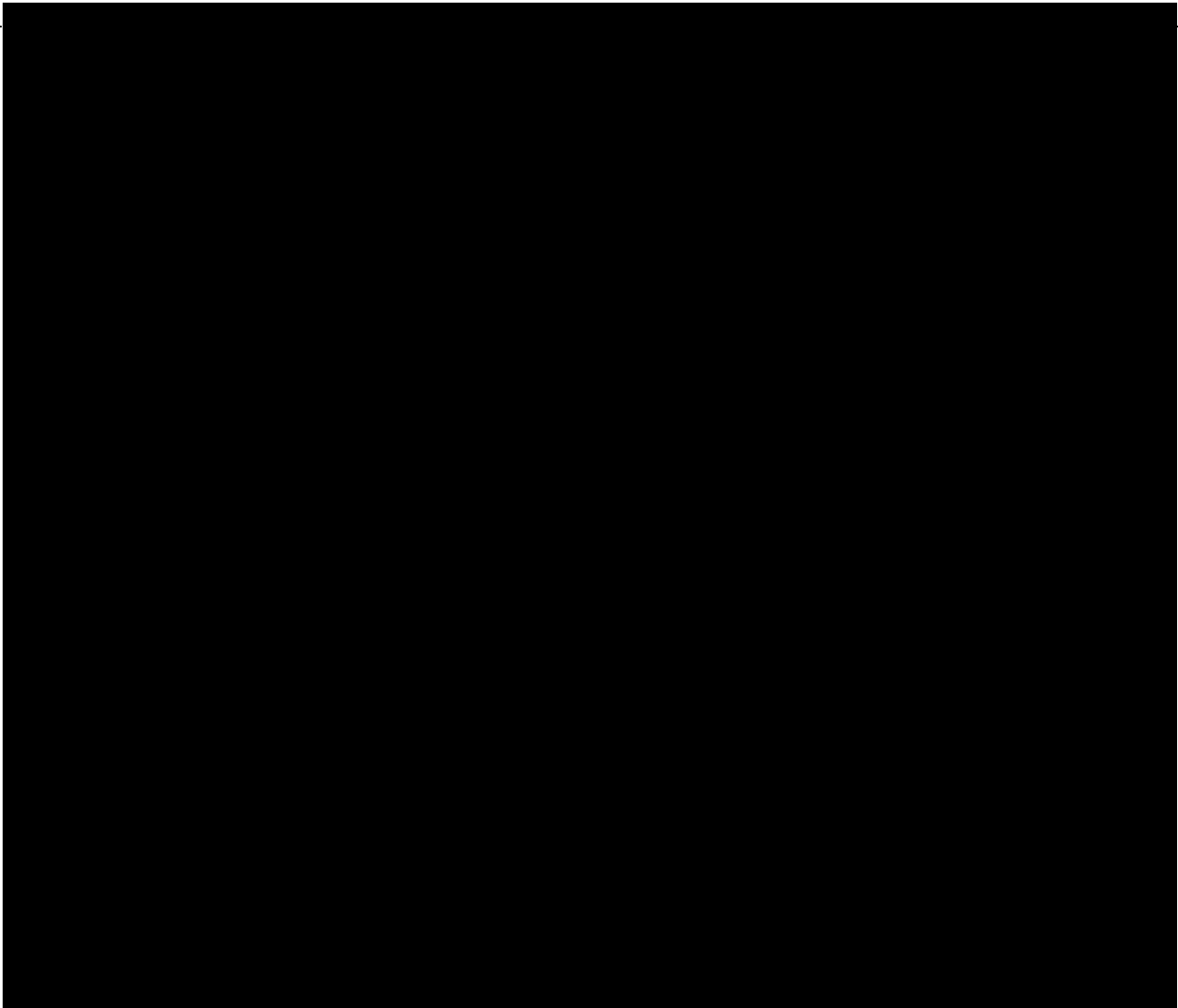


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Furthermore, [REDACTED]



Ex. CC-36 (Product Images) at 9 (showing Netlist DDR5 RDIMM as including SK hynix H5CG48AGB0 DDR5 SDRAM).



Ex. CC-36 (Product Images) at 18 (showing Netlist DDR5 MRDIMM as including SK hynix H5CG48BHB0 DDR5 SDRAM).

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Ex. CC-36 (Product Images) at 2 (showing Netlist DDR5 Overclock RDIMM as including SK hynix H5CG44AGB0 DDR5 SDRAM).

See also Ex. CC-53C (0887-99A-1M6060 Prelim 02-08-22) at 3

Ex. CC-54C (0888-97A-1A6B61-X00) at 3

).

And, based on SK hynix advertising that SK hynix DDR5 SDRAMs are JEDEC-compliant, therefore each of the Practicing '366 Products includes JEDEC-compliant DDR5 SDRAMs. See Ex. CC-72 (SK hynix DDR5) at 1:

Next-generation Memory

Having developed the industry's first JEDEC-compliant DDR5 DRAM, SK hynix expects to lead the era of next-generation memory with its 16Gb or 24Gb DDR5. The performance enhancements and TCO savings over DDR4 realize greater value for customers, while the 20% smaller power footprint delivers value for the environment and society as well.

Thereby, each of the Practicing '366 Products includes components that are configured to receive power by at least three regulated voltage lines (VDDQ, VDD, and VPP), as required by the JEDEC

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standard. *See* Ex. CC-55 (JESD79-5B v1.20) at 6 (annotated):

Symbol	Type	Function
LBDQ	Output	Loopback Data Output: The output of this device on the Loopback Output Select defined in MR53:OP[4:0]. When Loopback is enabled, it is in driver mode using the default RON described in the Loopback Function section. When Loopback is disabled, the pin is either terminated or HiZ based on MR36:OP[2:0].
LBDQS	Output	Loopback Data Strobe: This is a single ended strobe with the Rising edge-aligned with Loopback data edge, falling edge aligned with data center. When Loopback is enabled, it is in driver mode using the default RON described in the Loopback Function section. When Loopback is disabled, the pin is either terminated or HiZ based on MR36:OP[2:0].
RFU	Input/Output	Reserved for future use
NC		No Connect: No internal electrical connection is present.
VDDQ	Supply	DQ Power Supply: 1.1 V
VDD	Supply	Power Supply: 1.1 V
VSS	Supply	Ground
VPP	Supply	DRAM Activating Power Supply: 1.8V
ZQ	Reference	Reference Pin for ZQ calibration. This ball is tied to an external 240 ohm resistor(RZQ), which is tied to V _{SS} .

Therefore, Netlist DDR5 RDIMMs include SDRAM components that would be powered by VDD, VDDQ, and VPP. And, Netlist DDR5 RDIMMs include additional components, such as Montage M88DR5RCD02 RCD, which is configured to receive power from at least VDD and VDDIO regulated voltage lines. *See* Ex. CC-43 (M88DR5RCD02 Flyer) at 1 (“1.1V VDD and 1.0V VDDIO voltages”). Because the 1.1V is to support I/O communication with host and DRAM, it is more likely that the 1.1V is connected to the VDDQ rail than the VDD rail. But, that would not affect the analysis.

In Netlist MRDIMMs, the voltage rails for the SDRAMs would be the same because the same DDR5 SDRAMs are used. *See* Ex. CC-13 (JEDEC DDR5 MRDIMM Plan) at 1 (one planned feature for MRDIMM is “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption”). Netlist’s MRDIMMs also use Montage’s MRCD01. *See* images of Netlist MRDIMM in 1[pre]. Montage states that MRCD receives a 1.1V VDD and a 1.0V VDDIO input. *See* Ex. CC-12 (M88MR5RCD01 Flyer English) at 1 (“1.1V VDD and 1.0V VDDIO voltages”). Because the 1.1V is to support I/O communication with host and DRAM, it is more likely that the 1.1V is connected to the VDDQ rail than the VDD rail. But, that would not affect

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the analysis.

Some examples of combinations of components and regulated output voltages are as follows:

Components	Regulated Voltage
First Component: RCD (or MRCD)	VDDIO
Second Component: D1 SDRAM	VDD
Third Component: D2 SDRAM	VPP
Fourth Component: D3 SDRAM	VDDQ

or

Components	Regulated Voltage
First Component: D1 SDRAM	VDDQ (or VDD)
Second Component: RCD (or MRCD)	VDDIO
Third Component: D2 SDRAM	VPP
Fourth Component: D3 SDRAM	VDD (or VDDQ)

Components	Regulated Voltage
First Component: D1 SDRAM	VDD
Second Component: D2 SDRAM	VPP
Third Component: D3 SDRAM	VDDQ
Fourth Component: RCD (or MRCD)	VDDIO

or

[REDACTED]

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Components	Regulated Voltage
First Component: D1 SDRAM	VDD
Second Component: D2 SDRAM	VDDQ
Third Component: D3 SDRAM	VPP
Fourth Component: RCD (or MRCD)	VDDIO

or

First Component: D1 SDRAM	VPP
Second Component: D2 SDRAM	VDDQ
Third Component: D3 SDRAM	VDD
Fourth Component: RCD (or MRCD)	VDDIO

or

First Component: D1 SDRAM	VPP
Second Component: D2 SDRAM	VDD
Third Component: D3 SDRAM	VDDQ
Fourth Component: RCD (or MRCD)	VDDIO

or

First Component: D1 SDRAM	VDDQ
Second Component: D2 SDRAM	VPP
Third Component: D3 SDRAM	VDD
Fourth Component: RCD (or MRCD)	VDDIO

or

First Component: D1 SDRAM	VDDQ
Second Component: D2 SDRAM	VDD
Third Component: D3 SDRAM	VPP
Fourth Component: RCD (or MRCD)	VDDIO

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Additionally, a voltage regulator is expected to generate a regulated voltage (“V_{INTERNAL}”) needed to operate the circuits in the PMIC.

Components	Regulated Voltage
First Component: SPD	V _{DDSPD}
Second Component: TS	V _{DDIO}
Third Component: PMIC’s internal circuitry	V _{INTERNAL}
Fourth Component: RCD (or MRCD)	V _{DDQ}

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[1.d.2] wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines,

As also shown in the datasheets and schematics above, each component of the set of components is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines, as explained above and below.

For example, to the extent that the four regulated voltages are respectively VDD, VDDQ, VPP and VDDIO, the set of components and the associated regulated voltages are as follows:

Components	Regulated Voltage
RCD	VDDQ and VDDIO
SDRAMs	VDD, VDDQ, VPP

Each component of the set of components (RCD and SDRAMs) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, as explained above.

Note that in this combination, SPD and TS are not part of the set of the component, as they also derive power from VDDSPD, which is not one of the named four regulated voltages.

As another example, to the extent that the four regulated voltages are respectively VDDSPD, VDDQ, VDDIO and V_{Internal}, the set of components and the associated regulated voltages are as follows:

Components	Regulated Voltage
SPD	VDDQ, VDDIO
RCD (or MRCD)	VDDQ, VDDIO
TS	VDDSPD, VDDIO
PMIC internal circuitry	V _{Internal}

Each component of the set of components (SPD, TS, RCD and NVM/register in PMIC) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, as explained above.

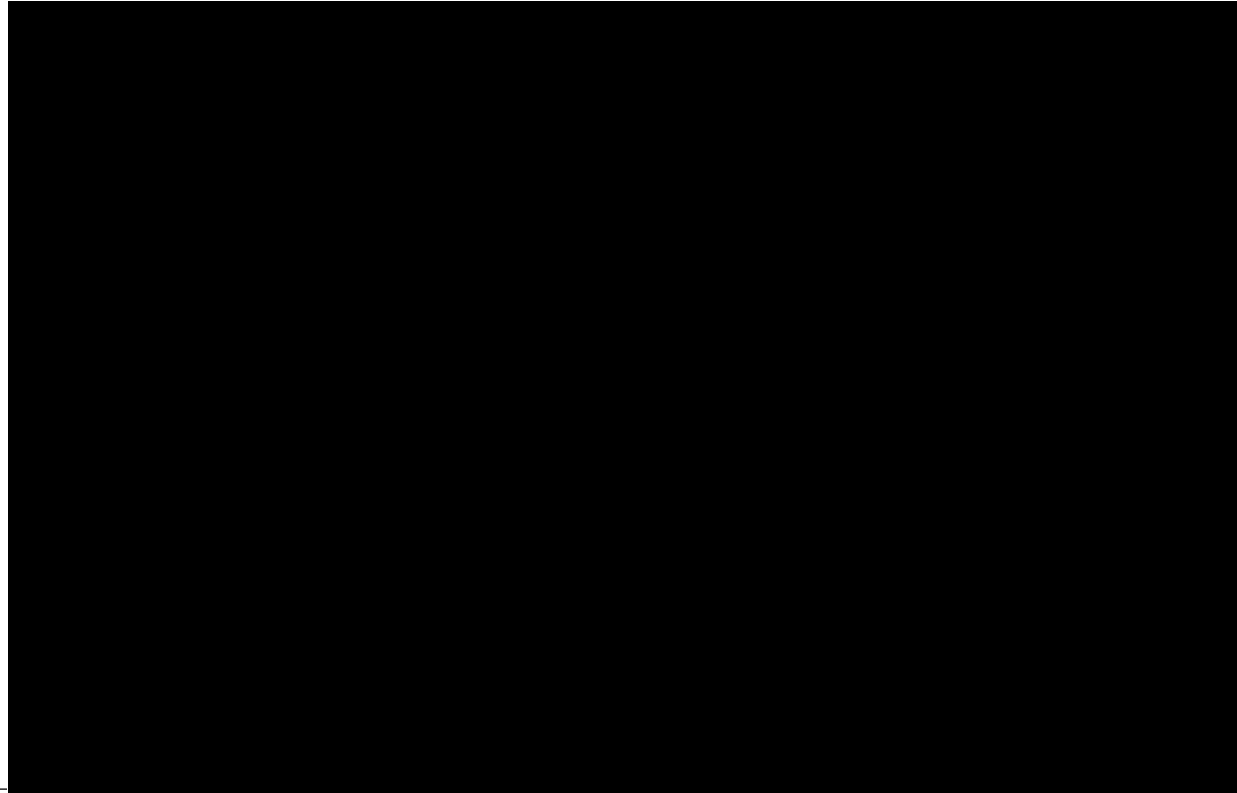
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[1.e] wherein the plurality of DDR SDRAM devices includes:
a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;
a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

The plurality of DDR SDRAM devices (*e.g.*, SDRAM devices in a rank) includes a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal (*e.g.*, those SDRAMs in channel A) and a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal (*e.g.*, those SDRAMs in channel B).

For example, Netlist's DDR5 VLP RDIMMs

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 1; *see also, e.g.*, Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9–10 (annotated):



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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1; *see also, e.g.*, Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10–12 (annotated):

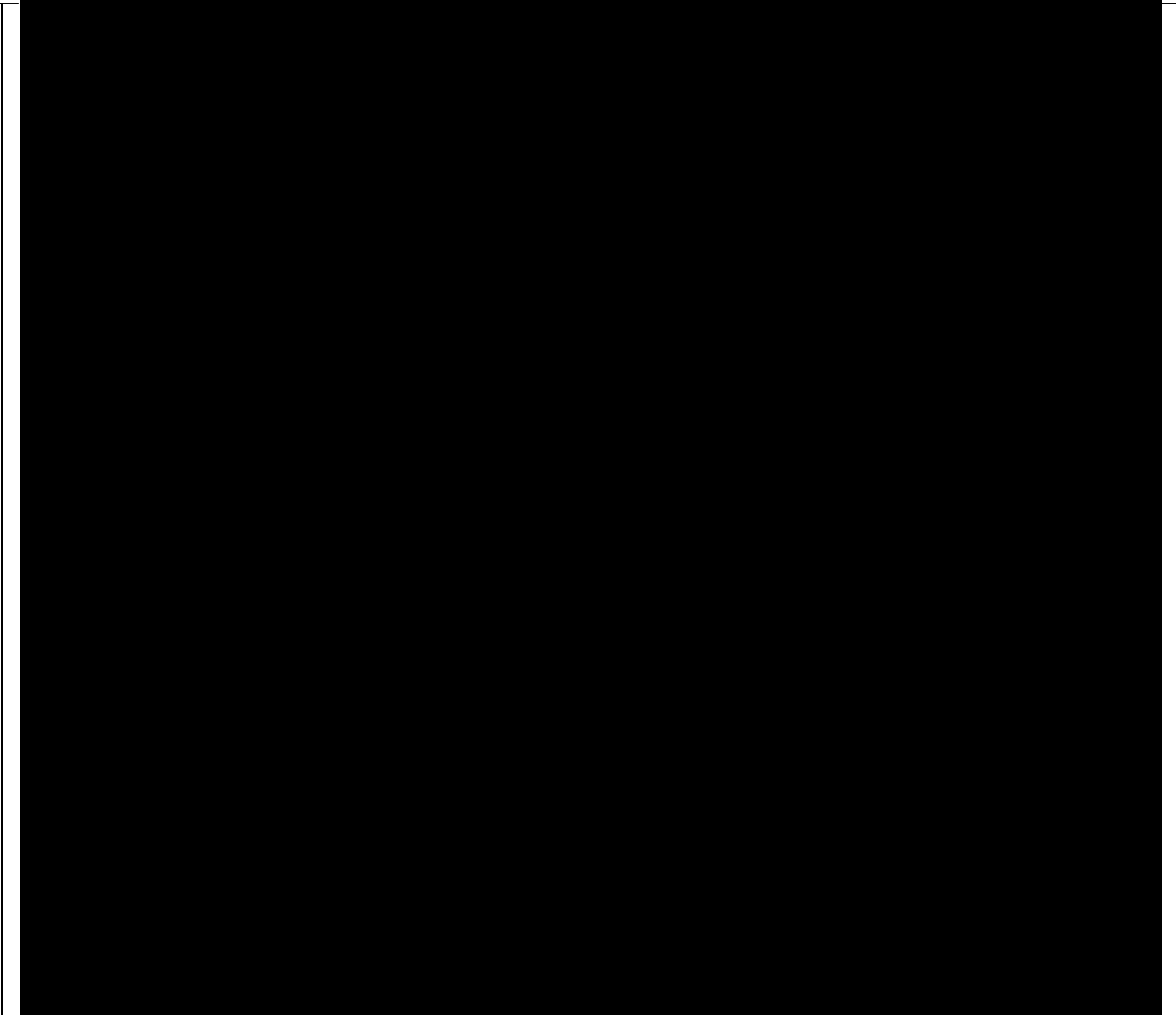
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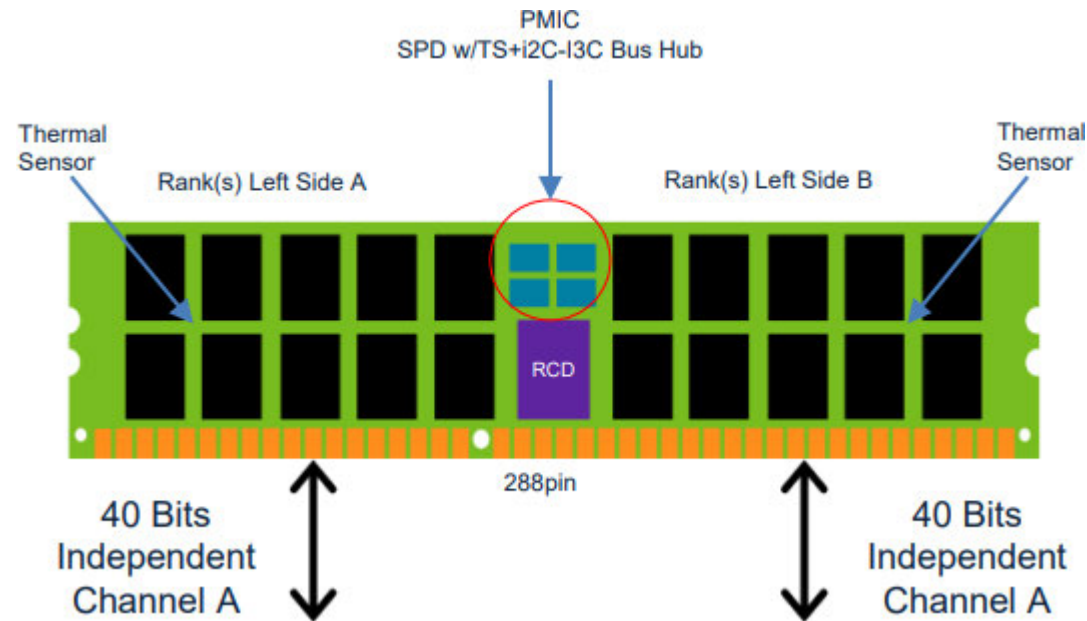


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	 As yet an additional example, Netlist advertises DIMMs, representative of Netlist DDR5 VLP RDIMMs,
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as including a Rank on each side of the board, (Left Side A and [Right] Side B). Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1:



Therefore, each side is a rank of 10 SDRAMs. Based on the specific configuration of Netlist DDR5 VLP RDIMM, as shown in the table, there may be as little as 10 SDRAMs, and, in such configurations, based on information and belief, one side and/or rank of the Netlist DDR5 VLP RDIMM includes at least 5 SDRAMs, and the other side and/or rank of the Netlist DDR5 VLP RDIMM includes at least 4 SDRAMs. See Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 2:

[REDACTED]

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NETLIST DDR5 RDIMM Ordering Information

DIMM Type	Form Factor	Height	Density	Config	Number of DRAM
Server RDIMM	LP	31.25	16GB	1Rx8	10
			32GB	1Rx4	20
			32GB	2Rx8	20
			48GB	1Rx4	20
			48GB	2Rx8	20
			64GB	2Rx4	20
			96GB	2Rx4	20
			128GB	4Rx4	40
	VLP	18.75	16GB	1Rx8	10
			32GB	2Rx8	20
			48GB	2Rx8	20

In an alternative example, such as in the 2Rx8 VLP RDIMM configurations, the Netlist DDR5 VLP RDIMM includes 20 SDRAMs, and, in such configurations, based on information and belief, one side and/or rank of the Netlist DDR5 VLP RDIMM includes at least 5 SDRAMs (e.g., 10), and the other side and/or rank of the Netlist DDR5 VLP RDIMM includes at least 4 SDRAMs (e.g., 10).

Additionally, in the image of Netlist's DDR5 MRDIMM, Netlist's DDR5 MRDIMM includes 20 SDRAMs, which can be divided into two ranks of 10 SDRAMs each, as shown below. *See* Ex. CC-36 (Product Images) at 16 (annotated):

Second Group

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[1.e.1] wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines;

the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least

Each of the Practicing '366 Products is designed to input or output data when it receives power from the power rails of the system board and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode.

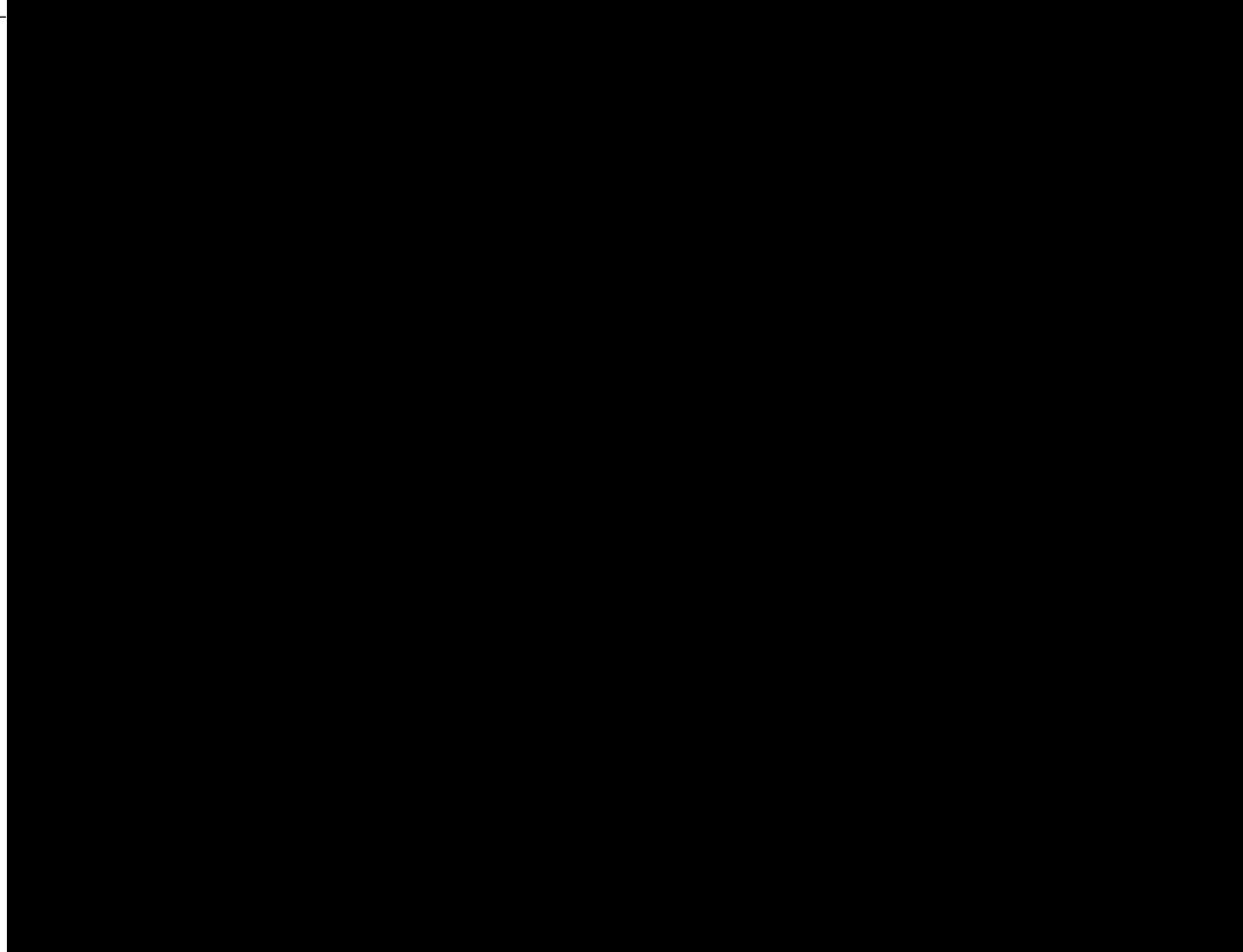
For example, as shown in *See also* Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 13 (

See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7–8 (annotated)



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eight 1-bit DDR error-
correcting code (ECC)
data signals; and



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[REDACTED]

[REDACTED]

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See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–6 (annotated)

([REDACTED]
[REDACTED]

[REDACTED]



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A self-refresh mode is a low-power mode where the DRAM periodically refreshes its content without commands from the memory controller. *See, e.g.*, Ex. CC-05 (JESD79-5C.01 v1.31) at 157 (stating that “[t]he clock is internally disabled (in the DRAM) during Self-Refresh Operation to save power” and that “[w]hile in self Refresh, the DDR5 SDRAM adjusts and updates its internal average periodic refresh interval”).

Furthermore, as explained in the JEDEC standard, DDR5 SDRAM devices will exit self-refresh mode when the CS signal input transitions from low to high (*e.g.*, when a read or write operation is initiated for a given channel, described below). *See* Ex. CC-05 (JESD79-5C.01 v1.31) at 157 (“[T]he DRAM will trigger Self-Refresh exit upon seeing the CS_n transition from low to high and stay high for

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tCSH_SRExit.”).

In a read or write operation, the first group of DDR SDRAM devices (*e.g.*, Rank0 or Rank1 SDRAMs in channel A) is configured to be enabled, in response to the first chip select signal (*e.g.*, DCS0_A_n or DCS1_A_n), to receive or output 40 1-bit DDR data signals in parallel via 40 data lines of the set of data lines (*e.g.*, via data lines associated with DQ[31:0]_A and CB[7:0]_A), independently of whether the second group of DDR SDRAM devices (Rank0 or Rank1 SDRAMs in channel B) is enabled, in response to the second chip select signal (*e.g.*, DCS0_A_n or DCS1_A_n), to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines (*e.g.*, via data lines associated with DQ[31:0]_B and CB[7:0]_B), because channel A and channel B receive independent C/A and CS signals as explained above. *See* evidence in 1.e.

See also, e.g., Ex. CC-05 (JESD79-5C.01 v1.31) at 117 (DDR5 SDRAM command truth table, showing that write or read operations which result in receiving or outputting data along the set of data lines are in response to combination of C/A signal):

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4 DDR5 SDRAM Command Description and Operation

4.1 Command Truth Table

- (a) Notes 1, 2, and 14 apply to the entire Command truth table
 (b) To improve command decode time, the table has been optimized to orient all 1-cycle commands together and all 2-cycle commands together, allowing CA1 to be used to identify the difference between a 1-cycle and a 2-cycle command.

[BG=Bank Group Address, BA=Bank Address, R=Row Address, C=Column Address, MRA=Mode Register Address, OP=Op Code, CID=Chip ID, CW=Control Word, X=Don't Care, V=Valid].

Table 30 — Command Truth Table

Function	Abbreviation	CS_n	CA Pins														NOTES
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 17, 20
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 11, 13, 20
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 13, 21, 20
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	WR Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	V	V	CID3	

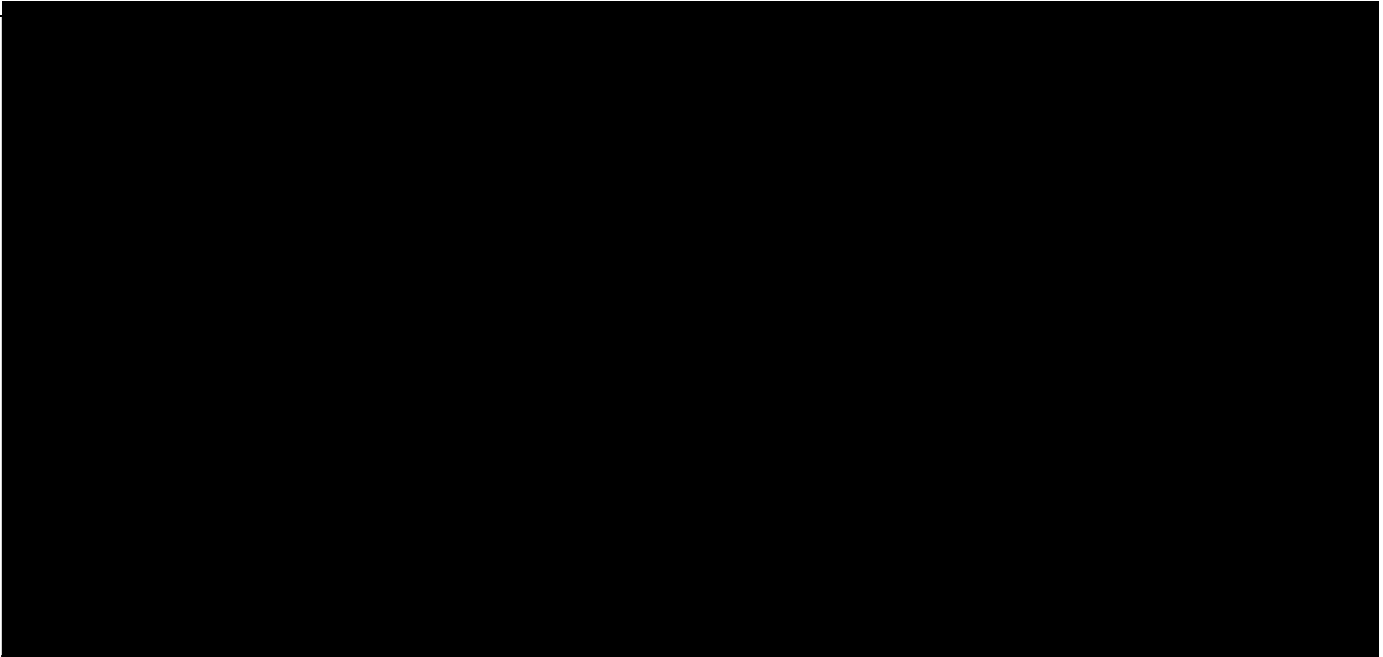
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See also:

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7 (annotated).

See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1, 4–6

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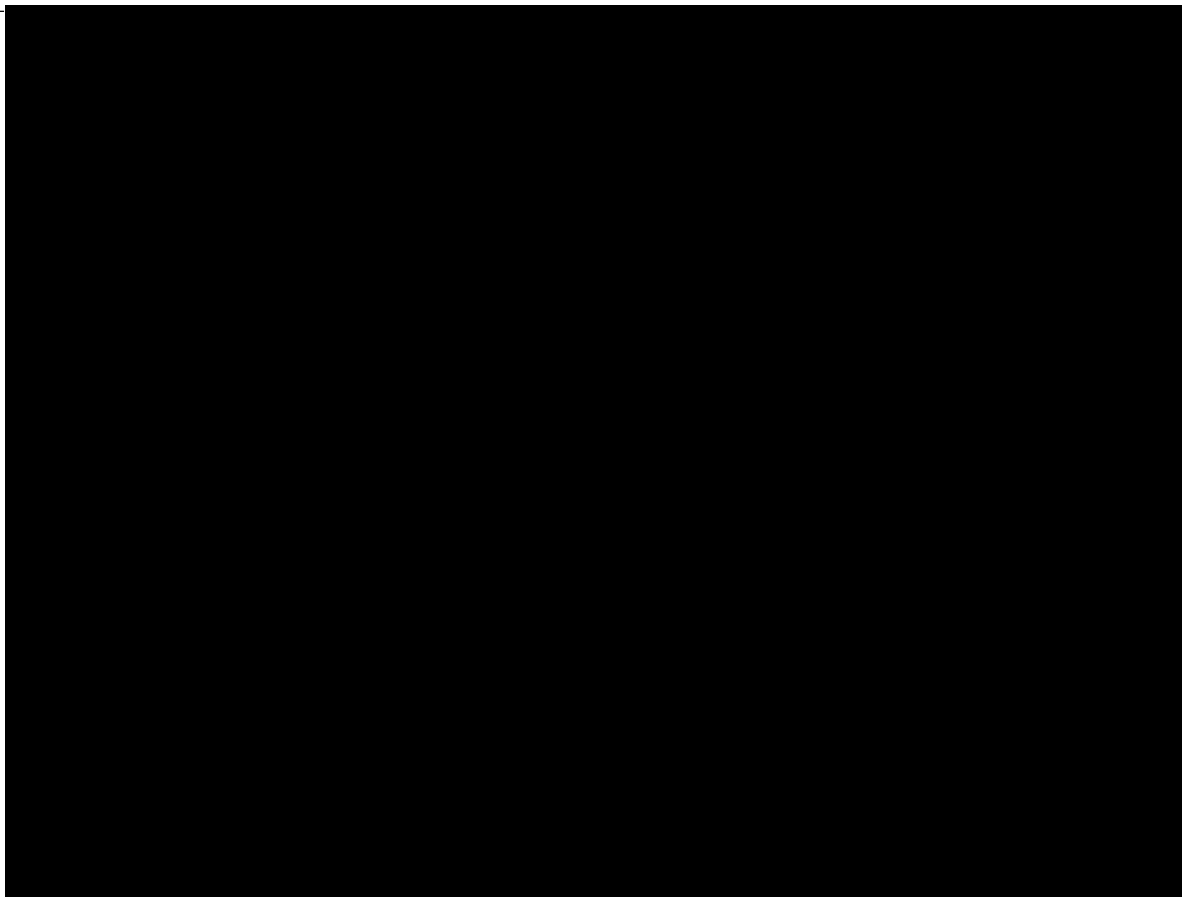


See also Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 10



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See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 11–12 (annotated)

[REDACTED]

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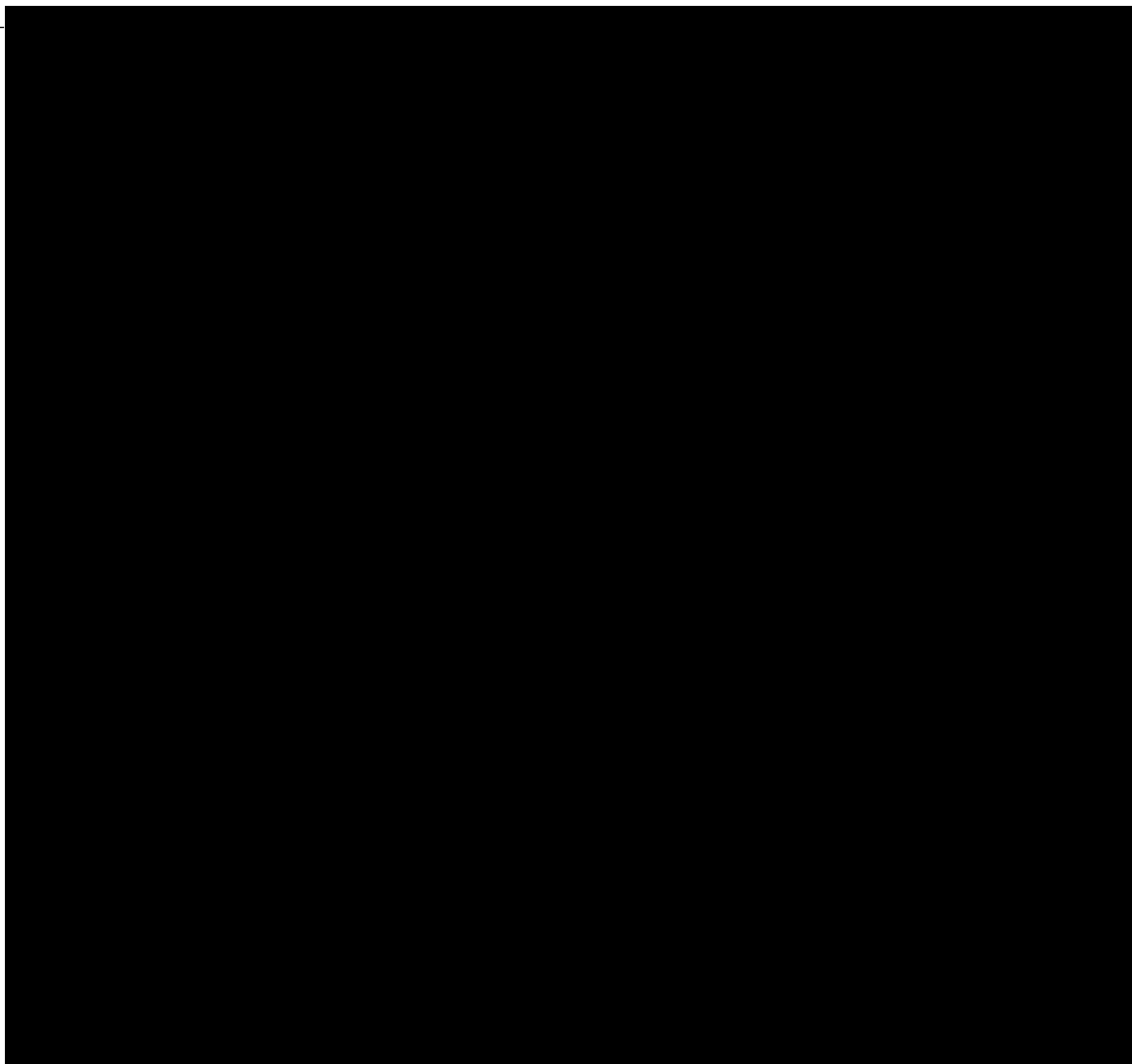
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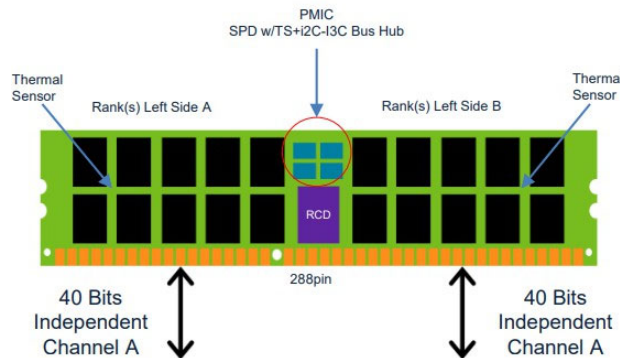
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See also Ex. CC-86 (Netlist DDR5 Overclock RDIMM) at 2 (advertising Netlist DDR5 Overclock RDIMMs as featuring a channel architecture of “40-bit data channel, 32 data + 8 ECC 2 channel per DIMM”); Ex. CC-57 (Lightning DDR5 High Performance Module Product Brief) at 1 (advertising the same as above).

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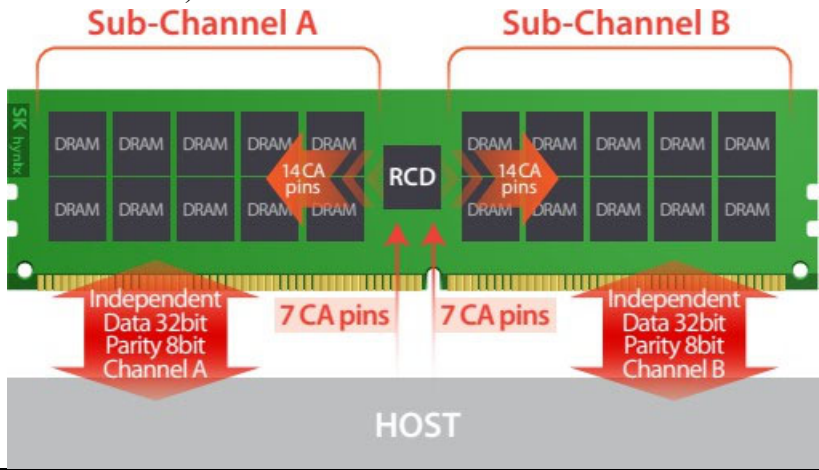
See also Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1 (showing 40 1-bit DDR data signals, including 32 1-bit DDR data signals and 8 1-bit DDR ECC signals, for each of the two independent channels):



DIMM Changes from DDR4 to DDR5

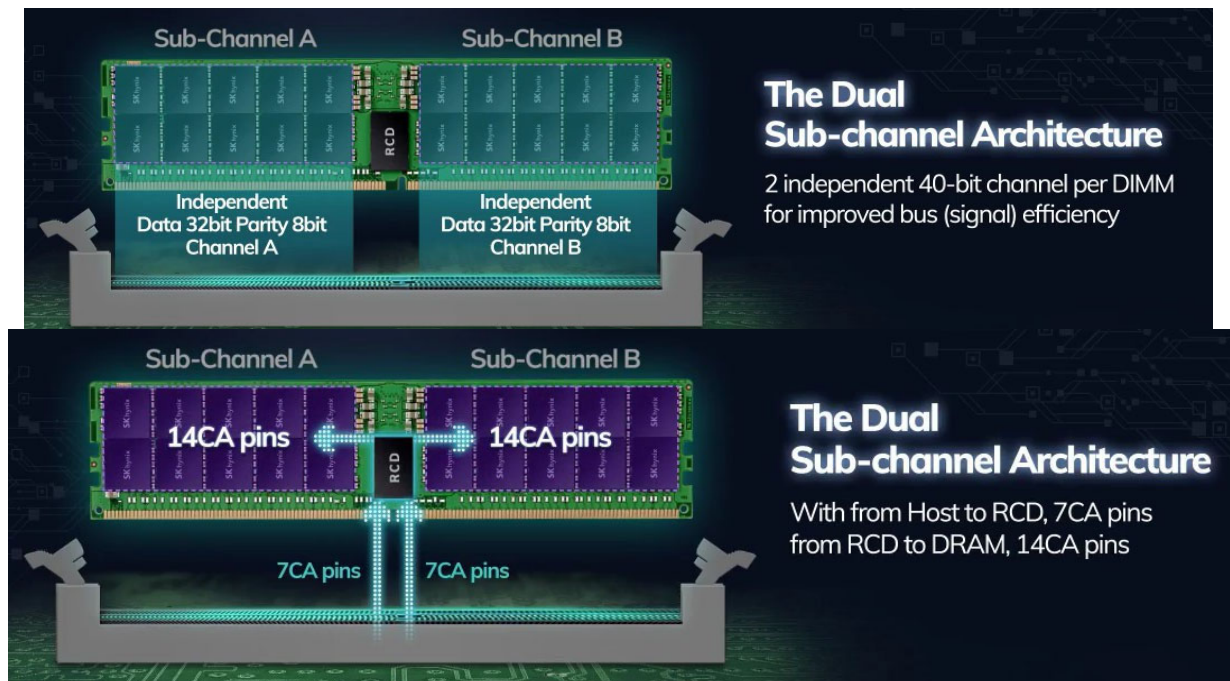
Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data+ 8 ECC 2 channel per DIMM
DfE Support	DB Only	RCD, DB, and DRAM

See also Ex. CC-40 (SK hynix A New Standard) at 5 (showing independent channels A and B for communicating 32 data bits and 8 parity bits between the respective sub-channels and the host and 7 distinct CA pins for each channel):



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See also Ex. CC-41 (RDIMM SK hynix) at 4–5 (showing the same as above):



Based on Netlist DDR5 MRDIMMs being backwards-compatible with existing JEDEC-compliant DDR5 RDIMM systems, as described in 1[pre], Netlist DDR5 MRDIMMs are capable and/or configured to enable at least one or more channels to receive or output 40 1-bit DDR data signals, independent of one or more other channels, wherein the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals. See 1[pre]; Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 2 (advertising the connector and physical layout for Netlist DDR5 MRDIMMs are the same as Netlist DDR5 RDIMMs, therefore Netlist DDR5 RDIMMs' connectors and physical layouts are representative of Netlist DDR5 MRDIMMs); Ex. CC-33 (Netlist DDR5 DIMMs) at 5 (advertising that Netlist's DDR5 MRDIMMs support "Backward Compatibility: Compatible with existing DDR5 RDIMMs, providing flexibility in system configurations").



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	Likewise, Netlist DDR5 MRDIMMs, include the same channel architecture as Netlist DDR5 RDIMMs and are therefore capable of enabling and/or configured to enable the aforementioned receiving or outputting for a first group of SDRAMs, based on a first chip select signal, and a second group of SDRAMs, based on a second chip select signal.
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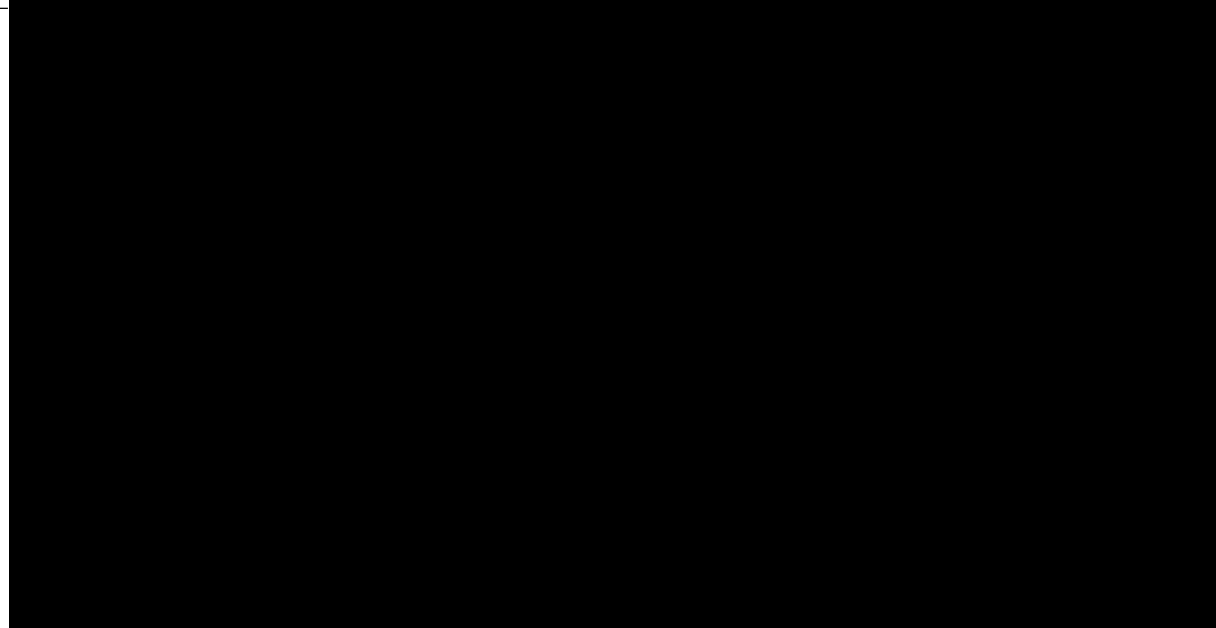
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[1.e.2] wherein the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

The sum (80 in Netlist's DDR5 VLP RDIMMs, DDR5 Lightning RDIMMs, DDR5 RDIMMs, and DDR5 MRDIMMs) of the at least 40 data lines and the integer number of data lines (40 in Netlist's DDR5 VLP RDIMMs, DDR5 Overclock RDIMMs, DDR5 RDIMMs, and DDR5 MRDIMMs) equals a total number of data conduits of the set of data conduits of the memory slot connector (40), and the total number of data conduits (80) is at least 72 data conduits.

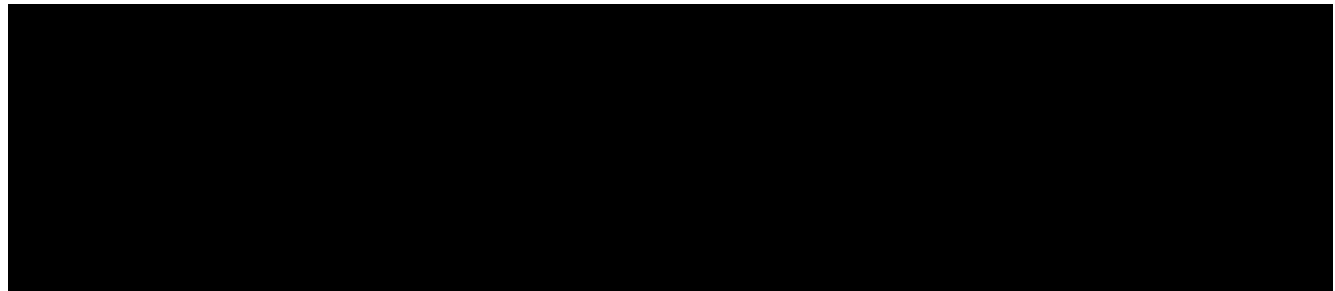
As shown above

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7 (annotated).

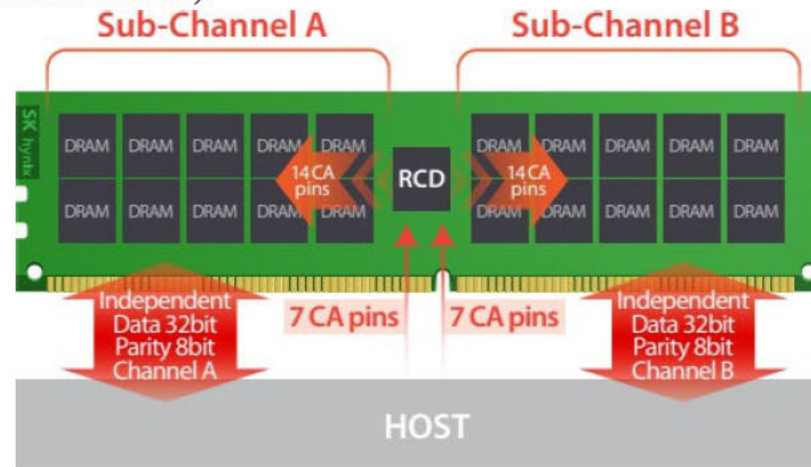
See also Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1, 4–6:



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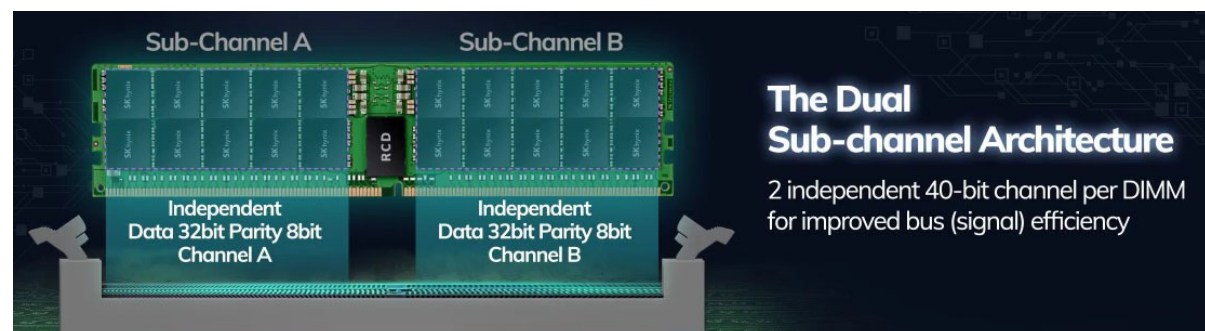
See also Ex. CC-86 (Netlist DDR5 Overclock RDIMM) at 2 (advertising Netlist DDR5 Overclock RDIMMs as featuring a channel architecture of “40-bit data channel, 32 data + 8 ECC 2 channel per DIMM”); Ex. CC-57 (Lightning DDR5 High Performance Module Product Brief) at 1 (advertising the same as above).

See also Ex. CC-40 (SK hynix A New Standard) at 5 (showing independent channels A and B for communicating 32 data bits and 8 parity bits between the respective sub-channels and the host and 7 distinct CA pins for each channel):



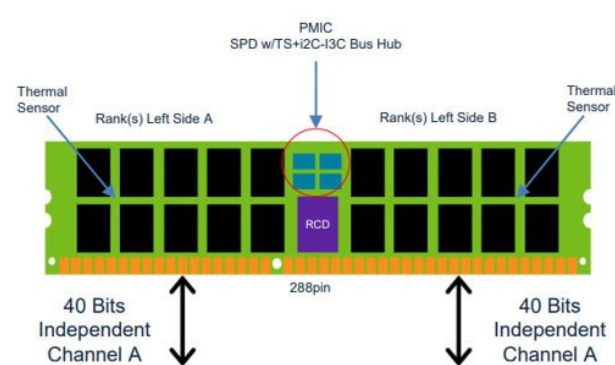
See also Ex. CC-41 (RDIMM SK hynix) at 5 (showing the same as above):

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See also Ex. CC-33 (Netlist DDR5 DIMMs) at 2 (advertising that DDR5 DIMMs “feature improved error correction capabilities, x80 vs. x72 with DDR4”).

Furthermore, each of DDR5 DIMMs’ 2 channels includes 40 data lines, comprised of 32 data lines and 8 ECC lines, summing to a total of 80 data lines. Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1 (annotated):



DIMM Changes from DDR4 to DDR5

Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data+ 8 ECC 2 channel per DIMM
DQE Support	DB Only	RCD, DB, and DRAM

Additionally, the connector and physical layout for Netlist DDR5 MRDIMMs are the same as Netlist DDR5 RDIMMs, therefore Netlist DDR5 RDIMMs’ connectors and physical layouts, as shown and described, are representative of Netlist DDR5 MRDIMMs. Ex. CC-71 (Netlist DDR5 MRDIMM

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Product Brief) at 2 (annotated):

DDR5 MRDIMM Features

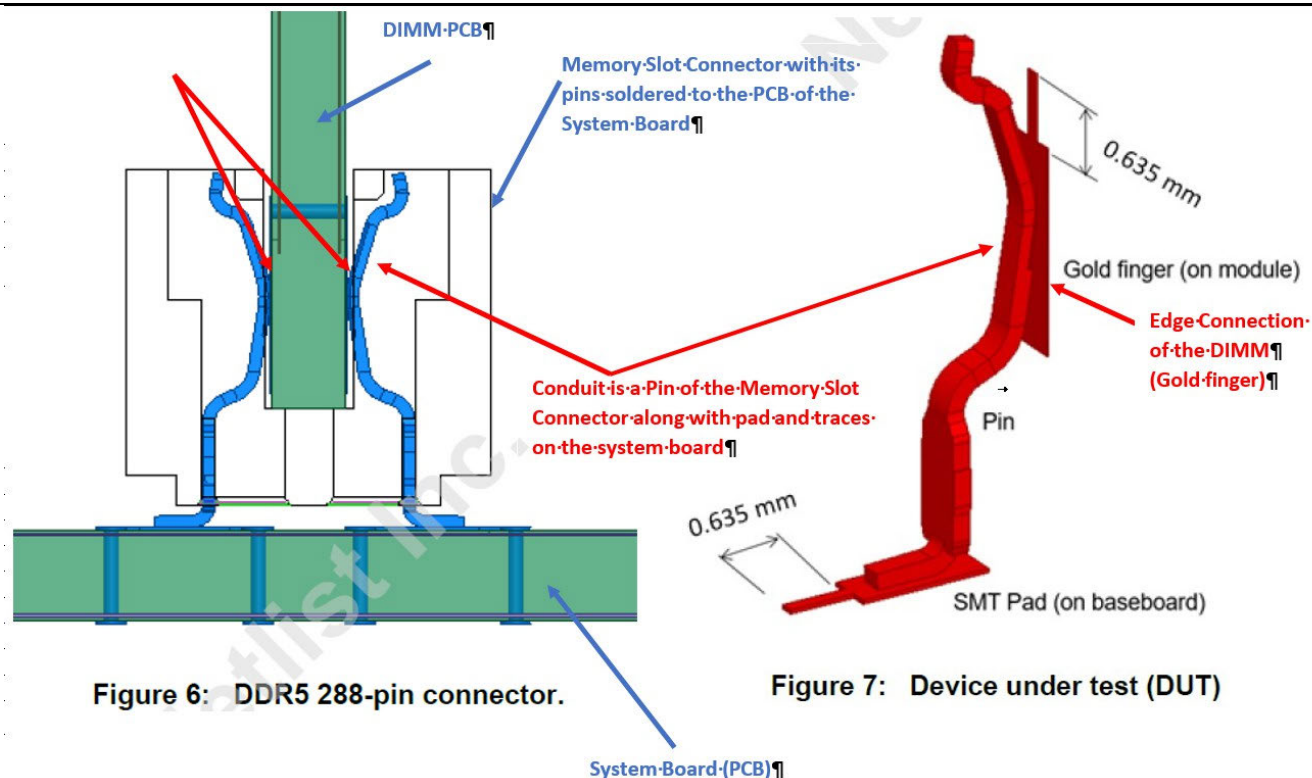
Features	Description
Bandwidth	Up to 8.8K MT/s, providing a 39% bandwidth upgrade
Latency	Up to 40% lower latency compared to traditional DDR5
Capacity	Available in sizes from 32GB to 128GB
Parallel DRAM Activation	Enables pairs of DRAM ranks to be activated and accessed in parallel
Multiplexed Data Streams	Doubles the data rate per signal pin
Power Management	Includes PMIC 5030 for efficient power management
Compatibility	Optimized for Intel Xeon 6 processors. They maintain the same connector and physical layout as RDIMMs, allowing easy integration into existing systems without modifications
Operating Voltage	1.1 volts
Operating Temperature	0°C to +70°C

See also Ex. CC-33 (Netlist DDR5 DIMMs) at 5 (advertising that Netlist’s DDR5 MRDIMMs support “Backward Compatibility: Compatible with existing DDR5 RDIMMs, providing flexibility in system configurations”).

Therefore, Netlist’s and SK hynix’s MRDIMMs also include at least 80 data lines.

Each edge connection is electrically coupled to a corresponding conduit in the memory slot, as illustrated below:

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Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6 and 7 (annotated) (showing an electrical connection between a connector in a memory slot and a corresponding edge connection of the DIMM).

Thus, the sum of the data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and given that there are 80 data lines per each of the Practicing '366 Products, that total number of data conduits (80) is at least 72 data conduits.